

Review

Small object detection: A comprehensive survey on challenges, techniques and real-world applications

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ABSTRACT

Small object detection (SOD) is a critical yet challenging task in computer vision, with applications like spanning surveillance, autonomous systems, medical imaging, and remote sensing. Unlike larger objects, small objects contain limited spatial and contextual information, making accurate detection difficult. Challenges such as low resolution, occlusion, background interference, and class imbalance further complicate the problem. This survey provides a comprehensive review of recent advancements in SOD using deep learning, focusing on articles published in Q1 journals during 2024–2025. We analyzed challenges, state-of-the-art techniques, datasets, evaluation metrics, and real-world applications. Recent advancements in deep learning have introduced innovative solutions, including multi-scale feature extraction, super-resolution (SR) techniques, attention mechanisms, and transformer-based architectures. Additionally, improvements in data augmentation, synthetic data generation, and transfer learning have addressed data scarcity and domain adaptation issues. Furthermore, emerging trends such as lightweight neural networks, knowledge distillation (KD), and self-supervised learning offer promising directions for improving detection efficiency, particularly in resource-constrained environments like Unmanned Aerial Vehicles (UAV)-based surveillance and edge computing. For example, several methods highlight the practical impact of architectural advances in SOD. FFCA-YOLO achieves APs of 0.748/0.617/0.909 on VEDAI/AI-TOD/USOD in remote sensing, and DsP-YOLO targets industrial defect detection, reaching 95.8% mAP on PCB-DET. We also review widely used datasets, along with standard evaluation metrics such as mean Average Precision (mAP) and size-specific AP scores. The survey highlights real-world applications, including traffic monitoring, maritime surveillance, industrial defect detection, and precision agriculture. Finally, we discuss open research challenges and future directions, emphasizing the need for robust domain adaptation techniques, better feature fusion strategies, and real-time performance optimization. By consolidating recent findings and identifying research gaps, this survey serves as a valuable resource for researchers aiming to advance SOD methodologies.

Abbreviation: AAFF, Adaptive Attention Feature Fusion; AFFM, Adaptive Feature Fusion Mechanism; AI-TOD, Artificial Intelligence for Tiny Object Detection; AP, Average Precision; BDFPN, Bidirectional Dense Feature Pyramid Network; Bi-AFPN-P2, Bidirectional Adaptive Feature Pyramid Network; BiFPN, Bidirectional Feature Pyramid Network; CGI, Computer-Generated Imagery; CV, Coefficient of Variation; DBB, Diverse Branch Block; DD-Head, Decoupled Detection Head; DETR, DETection TRansformer; DIOR, Dataset for Object Detection in Optical Remote Sensing Images; DOTA, Dataset for Object Detection in Aerial Images; DSCA, Depthwise Separable Channel Attention; EMA, Efficient Multi-scale Attention; FEM, Feature Extraction Module; FFN, Feature Fusion Network; FMA, Feature-Aware Module; FPN, Feature Pyramid Network; FPS, Frames Per Second; GAN, Generative Adversarial Networks; HDPCov, Halved Deep Pointwise Convolution; IoU, Intersection over Union; KD, Knowledge Distillation; LKSP, Large Kernel Spatial Pyramid Pooling; mAP, mean Average Precision; mAR, Mean Average Recall; MCSTA, Multi-head Channel and Spatial Trans-Attention; MCDM, Multi-Branch Dilated Convolution Module; MDFFAM, Multi-Directional Feature Fusion Attention Mechanism; MFFSODNet, Multiscale Feature Fusion Small Object Detection Network; MHA, Multi-Scale Hybrid Attention; MgD, Multi-Granularity Detection; MPDIoU, Minimum Point Distance Intersection over Union; MPM, Motion Processing Module; MSFEM, Multiscale Feature Extraction Module; MVA, Motion-Vision Alignment; NMS, Non-Maximum Suppression; PPN, Processed Pixel Number Percentage; PSFE, Point-Score Feature Extraction; RFRM, Receptive Field Refinement Module; SCFPN, Scale Compensation Feature Pyramid Network; SJR, Scientific Journal Rankings; SLAM, Simultaneous Localization and Mapping; SNR, Signal-to-Noise Ratio; SOIoU, Small Object Intersection over Union; SOD, Small Object Detection; SODA-D, SOD Dataset; SR, Super-Resolution; UAV, Unmanned Aerial Vehicle; UAVDT, Unmanned Aerial Vehicle Detection and Tracking; USV, Unmanned Surface Vehicle; VEDAI, Vehicle Detection in Aerial Imagery; VOV-GSCSP, View Group Shuffle Cross Stage Partial Network; WSODD, Water Surface Object Detection Dataset.

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1. Introduction

Small object detection (SOD) has emerged as a pivotal task in computer vision due to its significant role in various real-world scenarios. Unlike larger objects, small objects often contain limited spatial and contextual information, making their detection essential for applications where even minor elements can have a major impact. For instance, in safety-critical systems like autonomous vehicles, failing to detect small obstacles or pedestrians could result in catastrophic outcomes. The ability to accurately detect and localize small objects enhances the effectiveness and reliability of image-based decision-making systems, marking it as a key area of research (Cheng et al., 2023). Furthermore, advancements in this field contribute to the broader development of robust and scalable computer vision models that cater to diverse requirements in different domains.

The utility of SOD spans across a wide spectrum of applications, each critical to its respective field. In surveillance systems, accurately detecting small objects like unattended bags, individuals in crowds, or suspicious drones is crucial for maintaining public safety. In medical imaging, detecting small abnormalities such as tumors or micro-calcifications significantly aids early diagnosis and treatment, potentially saving lives (Feng et al., 2023). Similarly, in autonomous vehicles, recognizing small objects such as road signs, cyclists, or animals on the road ensures safe navigation. Moreover, in remote sensing, identifying small structures or vehicles in satellite imagery plays a vital role in urban planning and disaster response. These diverse applications underline the importance of developing highly accurate SOD methods tailored to meet domain-specific challenges.

Despite its importance, SOD remains a challenging task due to several inherent difficulties. The small size of the objects results in limited resolution and inadequate feature representation, making it difficult for models to differentiate them from the background. Moreover, small objects often appear in cluttered environments, where they may overlap with larger objects or be occluded by noise and artifacts, further complicating detection (Iqra and Javed, 2024). The class imbalance issue, where small objects are underrepresented in datasets, adds another layer of complexity, often leading to biased predictions. To address these challenges, researchers are exploring innovative techniques such as multi-scale feature extraction, super-resolution (SR) approaches, and attention mechanisms to enhance detection accuracy for small objects in various contexts.

The primary objective of this survey is to provide a comprehensive overview of recent advancements in SOD using deep learning, focusing on articles published in Q1 journals (based on Scientific Journal Rankings (SJR)) during 2024 and 2025 indexed in Scopus with "Small Object Detection" in their titles. The result of the search for the relevant articles in the Scopus database is a total of 76 papers in which the term "Small Object Detection" appears in the title. By reviewing these cutting-edge studies, we aim to analyze the challenges, techniques, datasets, evaluation metrics, and real-world applications related to the field. This narrative review consolidates current knowledge and delineates key research gaps and emerging trends to inform future studies. Compared to previous review articles (Cheng et al., 2023; Feng et al., 2023; Iqra and Javed, 2024), our work offers a more detailed and updated perspective by integrating insights from the most recent publications. Our paper provides a comprehensive review of the challenges and diverse methods in SOD, focusing on emerging application domains such as autonomous systems, remote sensing, medical imaging, and agriculture. It also delves into performance evaluation metrics and new benchmark datasets. By classifying methods and evaluating performance on key SOD tasks, this paper clearly and comprehensively outlines the place of these techniques in the SOD field. Furthermore, the paper discusses future research directions that have been less explored in other review articles, such as few-shot learning, weakly supervised learning, and model interpretability improvements. The contributions of the paper are as follows.

1. In this study, a comprehensive review of recent papers from 2024 to 2025 revealed gaps in research stemming from challenges in the architecture of proposed models for accurate SOD and the presence of small objects in images. This paper identifies the challenges small objects present in images, as well as the difficulties faced by proposed models. This paper identifies and categorizes the primary challenges and introduces them as research gaps, highlighting the need for further research and the development of new methods to address them.
2. This paper provides a comprehensive review of strategies for improving SOD performance from the following perspectives: (1) the application of recent trends in enhancing deep learning models' performance when faced with challenges posed by complex environments, (2) the impact of image quality, including SR, multi-scale information, and feature fusion across network layers, on detection accuracy, (3) the influence of data augmentation techniques and synthetic data on the performance of proposed models, and (4) the effect of advanced learning strategies, such as multi-task learning and transfer learning, in improving model adaptability in real-world scenarios.
3. In our review paper, we introduced ten key applications of SOD. These applications include remote sensing and aerial surveillance, Unmanned Aerial Vehicles (UAV) and drone applications, automated systems, industrial applications, monitoring and security, natural resource and environmental monitoring, medical imaging and biological analysis, marine and oceanic applications, intelligent transportation systems, and specialized applications in agriculture and forestry. Introducing these ten domains in a comprehensive and analytical manner allows for a deeper exploration of the challenges and research opportunities in each of these areas, and can contribute to the development of more effective methods and techniques in SOD.
4. This study offers a structured overview of SOD applications across domains like autonomous systems, environmental monitoring, medical imaging, and smart agriculture. The review is notable for its broad coverage, innovative categorization, and use of up-to-date sources.

To structure the discussion effectively, the paper is organized as follows: [Section 2](#) introduces the definitions and background of SOD, including its distinction from general object detection. [Section 3](#) delves into the major challenges, including resolution limitations, background noise, small object size, and dataset imbalances. [Section 4](#) provides an in-depth examination of advanced deep learning techniques and Recent trend, exploring innovations in neural network architectures, SR methods, data augmentation, transfer learning, and problem-specific approaches. [Section 5](#) reviews key datasets like COCO, Pascal VOC, DOTA, and VisDrone, alongside evaluation metrics such as mAP, precision, recall, and F1-score, with an emphasis on their relevance to SOD. [Section 6](#) outlines real-world applications, spanning areas like satellite imagery, video surveillance, medical imaging, and agriculture. Finally, [Section 7](#) analyzes the future directions in this field. The paper concludes with a summary of findings and implications in [Section 8](#). This comprehensive structure ensures that the survey not only serves as a reference for researchers but also facilitates the development of innovative solutions in SOD. The main contributions of the paper are presented below.

2. Definitions and background

Small objects are typically characterized by their limited size, occupying a minimal number of pixels in an image, which makes their detection particularly challenging. The definition of small objects often depends on the dataset, application domain, and contextual criteria used in the analysis. Below, we explore various perspectives and definitions based on prominent datasets and research works.

2.1. Pixel-based definitions

A common approach to defining small objects is through their pixel dimensions or areas. For example, the MS COCO dataset, a widely-used benchmark in object detection, categorizes small objects as those with an absolute area smaller than 32×32 pixels, equivalent to less than 1024 square pixels. This criterion is a standard reference point in many studies, as seen in the works utilizing MS COCO-based evaluations (Z. Chen et al., 2024; S. Chen et al., 2024; Tian et al., 2024).

In the context of satellite and aerial imagery, small objects can appear even smaller. For instance, moving objects in satellite videos are often less than 20×20 pixels, which highlights the increased detection difficulty due to their diminutive size (S. Chen et al., 2024; Lei and Liu, 2024). Similarly, small objects in UAV imagery are frequently defined as those occupying a limited number of pixels (e.g., fewer than 32×32) and often consist of sparse, low-resolution features (R. Wang et al., 2024; Cao et al., 2024).

2.2. Relative size criteria

Some definitions use the relative size of an object compared to the entire image area. For instance, objects occupying less than 1 % of the image area are categorized as small (Liu and Jiang, 2024; Tong and Wu, 2024; Yang et al., 2024; L. Ni et al., 2024). This criterion becomes particularly relevant in high-resolution images where objects might have sufficient pixel counts but still appear small relative to the overall image dimensions.

Other studies define "small objects" using dynamic thresholds based on the dataset or application. For instance, in the TT-100 K dataset, small objects are described as those smaller than 30×30 pixels in a 2048×2048 image, which corresponds to less than 1.5 % of the total image resolution (Zhu et al., 2024). This relative approach provides flexibility in adapting the definition to diverse imaging conditions.

2.3. Categorization of small objects

Several studies go beyond binary definitions to classify small objects into multiple subcategories:

- Tiny Objects: Defined as having areas up to 4500 square pixels.

- Small Objects: Larger than tiny objects but below specific thresholds based on datasets like MS COCO (Tong and Wu, 2024).
- Dense Small Objects (DSO): Overlapping objects in crowded scenes, such as urban areas captured by UAVs (C. Chen et al., 2024).

Table 1 outlines the challenges and features associated with detecting small objects in various contexts. Each row is dedicated to a specific category of small objects, highlighting their applications in fields such as video surveillance, robotics, and industrial environments.

2.4. Challenges highlighted by definitions

While definitions provide a framework for understanding SOD, numerous studies underscore the inherent difficulties in accurately detecting and classifying small objects. These challenges stem from both the intrinsic properties of small objects and the complexities of the environments in which they are found. Key challenges discussed in the literature include (F. Feng et al., 2025; Shao et al., 2024; Feng et al., 2024):

- Low Pixel Count: Small objects typically occupy fewer pixels in an image, resulting in limited feature representation. This reduction in pixel density makes it harder for traditional object detection algorithms to capture sufficient details for accurate classification and recognition. As the resolution of the object diminishes, the features that distinguish it become less prominent, contributing to lower detection performance.
- Background Interference: Due to their small size, these objects are often difficult to distinguish from complex and cluttered backgrounds. The limited spatial coverage of small objects makes them prone to blending with surrounding elements, especially in environments with high visual noise or dynamic backgrounds. This challenge is particularly evident in natural scenes, urban environments, or when small objects are partially occluded by other objects.
- Scale Variability: Small objects can undergo significant changes in appearance depending on their relative position, distance from the camera, and the angle of view. As perspective shifts, small objects may appear larger or smaller, or even be lost from view entirely. Variations in scale across different images or frames of video further complicate the detection task, as the model must adapt to recognize objects in diverse conditions without losing accuracy.

Table 1
Categorization and comparison of small objects in SOD.

References	Category	Applications and Examples	Challenges	Features
MICPL (S. Chen et al., 2024), ESOD (K. Liu et al., 2025)	Static Small Objects	Traffic signs, safety equipment and barriers, fixed industrial parts in production lines	Need for very high accuracy in identifying small details, Sensitivity to background noise and lighting conditions	Small and fixed size in the environment, Typically stationary or without positional change
ESOD (K. Liu et al., 2025), Libra-SOD (Zhou and Zhu, 2024), DNTFE-Net (B. Zhao et al., 2024)	Moving Small Objects	Pedestrians, small animals Moving objects in robotics and drones Small objects in video surveillance	Occlusion by other objects, Sudden changes in position and size, Motion blur	Small-sized objects that are moving and changing positions, Quick changes in viewpoint and scale
DNTFE-Net (B. Zhao et al., 2024)	Semi-transparent or Undefined Small Objects	Smoke and fog in industrial environments, Semi-transparent glasses and sprinklers, Hidden or invisible objects in security and surveillance	Difficult to extract stable and accurate features, Interference from variable and noisy backgrounds	Objects that are semi-transparent or undefined due to environmental conditions or nature of the object, Lack of full clarity
MICPL (S. Chen et al., 2024), Libra-SOD (Zhou and Zhu, 2024)	Small Objects in Complex and Crowded Environments	Small object identification in aerial and satellite images Object detection in markets and busy urban environments	Difficulty in separating objects from dense and complex backgrounds, Decreased accuracy due to numerous interferences	Positioned in detailed and diverse backgrounds, Large number of objects with high density
ESOD (K. Liu et al., 2025), Libra-SOD (Zhou and Zhu, 2024)	Small Objects with Scale and Lighting Variations	Small object detection in live videos with varying light, Applications in autonomous vehicles and robotics	Instability of features under severe lighting and shadow variations, Challenges in adapting the model to scale changes	Size and lighting that vary over time, Inhomogeneous lighting conditions and shadows

In addition to these core challenges, studies also highlight issues such as:

- **Occlusion:** Small objects are more likely to be partially or fully occluded by other objects in the environment. This makes it challenging to identify the object's true boundary and distinguish it from the occluding elements.
- **Motion Blur:** Rapid movement of small objects can introduce motion blur, resulting in unclear edges and degraded visual quality. This diminishes the effectiveness of feature extraction and poses challenges for accurate detection.
- **Low Contrast:** In many real-world scenarios, small objects might be poorly lit or placed in low-contrast environments, further hindering the detection capabilities of algorithms. In some cases, the object

may appear too similar to its background, making it challenging to identify.

These challenges are critical for researchers and practitioners to address, as they not only affect the performance of SOD models but also influence the selection of appropriate methodologies and the need for advanced features such as multi-scale detection, context-aware algorithms, and robust training techniques.

3. Challenges in SOD

Detecting small objects in images presents significant challenges across various fields, including industrial defect detection, surveillance, medical imaging, and remote sensing. These challenges arise due to a



Fig. 1. Key challenges in SOD and related issues.

variety of factors, such as limited appearance information, occlusion, background interference, and the inherent limitations of existing detection models. This section explores the key challenges in SOD (Fig. 1) and provides insights into current research and potential solutions.

The purpose of Fig. 1 is to allow the reader to easily understand the key challenges in SOD and recognize how these challenges are subdivided into different categories. This comprehensive classification in Fig. 1 provides a clear roadmap for understanding the challenges in SOD and highlights the need for specialized methods and architectures to address these issues.

3.1. Limited appearance information and occlusion

Small objects often lack sufficient distinctive visual features that allow them to be differentiated from the background or other similar objects. This lack of identifiable characteristics, combined with frequent occlusion, makes detecting small objects difficult. For instance, in drone-captured images, the small targets may suffer from poor resolution, feature indistinguishability, and occlusion, which all significantly impair detection performance. Additionally, small objects tend to be more susceptible to various environmental factors, including lighting interference, blur, and occlusion. In scenarios such as coal mines, SOD is hindered by challenging environmental factors like low illumination, high dust, and water vapor density (R. Jing et al., 2024; J. Wu et al., 2024; T. Aibibu et al., 2023; Ma et al., 2025).

Small objects often lack distinctive visual features, such as sharp edges, textures, colors, or patterns, which are crucial for differentiating them from the background or similar objects. Additionally, due to their size, they frequently suffer from poor resolution, making their features less discernible and harder to identify accurately. This lack of distinctive features and high resolution (R. Jing et al., 2024; Ma et al., 2025) leads to several challenges in detection. One major issue is feature indistinguishability, where small objects become difficult to distinguish from the background or other objects, particularly in cluttered environments. They are also highly susceptible to occlusion (J. Wu et al., 2024; Ma et al., 2025), as they are frequently blocked by other objects, further complicating detection efforts. Moreover, small objects are more vulnerable to environmental factors such as lighting changes, motion blur, dust, low illumination (e.g., in coal mines), and water vapor, all of which degrade detection accuracy (T. Aibibu et al., 2023).

These challenges significantly impair the performance of SOD systems in various applications. For instance, in drone imagery, issues like poor resolution and occlusion often result in missed or false detections. Similarly, in industrial settings such as coal mines, low light conditions and airborne particles exacerbate detection failures. Collectively, these factors hinder the reliability and effectiveness of SOD systems in real-world scenarios.

3.2. Challenges in localization and scale variation

Locating small objects is inherently more difficult due to the larger number of possible locations and the need for higher localization accuracy. Small objects often face challenges in achieving high Intersection-over-Union (IoU) values with anchor boxes, making precise localization harder (Zhou and Zhu, 2024). The small-scale nature of these objects requires that detectors exhibit an exceptionally fine-grained understanding of the image to accurately locate them. This is especially difficult when objects vary greatly in size and scale, leading to issues such as misalignment of the receptive field (R. Jing et al., 2024; T. Li et al., 2024) or mismatched anchor boxes (S. Li et al., 2024). In environments like coal mines, small objects often lack high-resolution data, and detection methods that rely on centralized cloud computing experience significant latency. This is exacerbated by concept drift, where the characteristics of data change over time, which degrades the detection accuracy of small objects (J. Wu et al., 2024).

Locating small objects presents several key challenges, particularly in terms of localization difficulty and scale-related issues. One major challenge is the high spatial uncertainty, as the large number of possible locations for small objects increases the complexity of pinpointing them accurately (Zhou and Zhu, 2024). Additionally, achieving precise localization is inherently difficult due to the small size of these objects. Detectors require higher localization accuracy, but obtaining high IoU values with anchor boxes is often challenging, leading to imprecise bounding box predictions (S. Li et al., 2024). Furthermore, detecting small objects demands a fine-grained understanding of images, requiring detectors to analyze details at an exceptionally high level. This level of detail is computationally intensive and adds to the complexity of the task (S. Chen et al., 2024; Sun et al., 2024).

Scale and structural issues further complicate the detection process. Small objects often exhibit significant variability in size and scale, which can cause mismatches between anchor boxes and the actual dimensions of the objects (R. Jing et al., 2024; D. Chen et al., 2024). This variability also leads to misalignment of the model's receptive field (the area "seen" by the detector) with the object's actual location or scale. Such misalignment reduces the reliability of localization, as the detector may fail to focus on the correct area or scale, impacting overall accuracy.

Environmental and technical problems also pose significant hurdles. In environments like coal mines, low-resolution data is a common issue, as small objects often lack high-quality imagery, further degrading detection performance (G. Song et al., 2024). Additionally, cloud-based detection methods, which rely on centralized computing, often suffer from latency, making real-time detection impractical. Dynamic data challenges, such as concept drift, further exacerbate the problem (J. Wu et al., 2024). Changes in data characteristics over time, such as shifts in dust levels, lighting conditions, or object appearance, can reduce detection accuracy. Models trained on older data may become outdated, struggling to adapt to these evolving conditions, which ultimately impacts their effectiveness in real-world scenarios.

3.3. Inefficiency in feature learning and background interference

The downsampling of input images in convolutional networks leads to a significant reduction in discriminative information, such as textures and edges. This reduction adversely affects the classification accuracy for small objects (R. Wang et al., 2024), (S. Li et al., 2024). Additionally, small objects are often overshadowed by dense background interference, leading to a low Signal-to-Noise Ratio (SNR). In the case of industrial defect detection or UAV surveillance, small objects are frequently embedded in complex, noisy backgrounds, making them even harder to identify (D. Zhao et al., 2024). For example, in industrial settings, machinery defects may be surrounded by intricate visual data, while in aerial surveillance, small targets are frequently obscured by irrelevant background details. These complex backgrounds make it significantly harder to isolate small objects from the surrounding visual noise, reducing the effectiveness of detection systems in real-world applications (S. Chen et al., 2024; Y. Zhang et al., 2024).

Structural limitations of detection networks, such as Feature Pyramid Networks (FPNs), also contribute to the challenge. These networks face a trade-off between shallow and deep layers. Shallow layers provide strong localization accuracy but lack rich semantic information, while deeper layers contain valuable semantic details but lose precise localization capabilities. This imbalance in performance leads to suboptimal detection (Zheng et al., 2024; Yuan et al., 2019), especially in environments where objects vary significantly in size, scale, or background complexity. As a result, detection networks struggle to achieve both accurate localization and meaningful semantic understanding simultaneously (F. Zhao et al., 2024).

3.4. Limitations of popular detection methods

Existing object detection methods, such as those based on CNNs,

were primarily designed for detecting larger objects and are thus not well-suited for SOD. These models often fail to preserve fine-grained details crucial for SOD. Popular detectors are also highly computationally intensive, particularly when working with high-resolution images, which significantly increases the resource demand and processing time (J. Liu et al., 2024).

Additionally, when training these models, there is often an imbalanced contribution between tasks in multi-task learning setups. This imbalance can hinder the model's ability to learn effective features for SOD, resulting in limited accuracy (S. Li et al., 2024). Furthermore, retraining existing methods to adapt them for SOD often requires significant modifications to the architecture, training process, or loss functions, which can be computationally expensive (C. Chen et al., 2024).

3.5. High computational costs and hardware resource limitations

One of the biggest obstacles to SOD is the high computational overhead associated with processing large datasets, especially when high-resolution images are required. SOD in environments such as aerial imaging with UAVs introduces additional challenges, including limited computing resources on embedded devices and the need for efficient real-time analysis (Z. Chen et al., 2024; G. Song et al., 2024; J. Liu et al., 2024; R. Jing et al., 2024). For instance, small-scale objects captured by drones often suffer from a lack of detailed information, and varying camera angles can lead to significant differences in object scale and density within a scene (Z. Chen et al., 2024).

Moreover, embedded devices on unmanned surface vehicles (USVs) or in coal mines have limited computational power, making it difficult to deploy large, complex perception models. This is further complicated by the necessity for these systems to adapt to new object categories during missions, especially when dealing with limited samples (few-shot learning). Existing algorithms that perform well on SOD are often too resource-intensive and unsuitable for edge computing scenarios, where latency and real-time responsiveness are critical (Liu and Jiang, 2024; R. Wang et al., 2024; W. Wang et al., 2024; Y. Gao et al., 2024). Two real-world deployments that address these challenges include a lightweight SOD method based on multilayer coordination and federated intelligence for coal mine IoT, which utilizes optimized backbone architectures and advanced federated learning strategies to reduce complexity and enhance detection accuracy (J. Wu et al., 2024), and a high-accuracy, real-time two-stage SOD model on FPGA, which adopts a low-resolution localization and high-resolution classification strategy for efficient hierarchical feature processing and improved performance under tight computational (S. Li et al., 2024).

3.6. Inconsistent performance across different scales and datasets

Another significant challenge is the performance gap between small and large object detection. This gap becomes even more exacerbated when the training and testing datasets differ substantially in terms of object scale. Small objects often fall outside predefined anchor or grid regions, leading to a loss of training data. In addition, the continuous convolutional processes in advanced networks tend to cause feature disappearance, further hindering accurate detection (Tong and Wu, 2024; L. Zhou et al., 2024).

Existing methods often struggle when small objects are densely packed, as these objects may have limited or no distinguishing features in the image. This results in performance degradation when object scales vary dramatically between training and testing phases. The lack of specialized datasets focusing on small objects in large, complex areas, such as aerial images (Jiang et al., 2024) or large-scale industrial settings (Zou et al., 2024), exacerbates this issue.

3.7. Solutions and emerging approaches

In response to these challenges, several innovative approaches have been proposed. One promising solution is the introduction of motion-inspired cross-pattern learning, which enhances detection by incorporating both motion and visual cues to overcome the challenges of SOD in dynamic environments. Additionally, Multi-Granularity Detection (MgD) (D. Chen et al., 2024) frameworks have been introduced to address issues like poor detection performance and sample imbalance, allowing for more accurate and robust detection of small objects across different scales.

Furthermore, recent developments in neural networks, such as enhancing FPNs (Tian et al., 2024) to better fuse low-level and high-level features, offer a potential solution to the semantic gap that often affects SOD. These advancements aim to preserve crucial localization information while enriching semantic understanding, thus improving the accuracy of SOD.

3.8. The impact of input noise and disturbances on SOD

SOD models are highly vulnerable to various input noise and disturbances. Key challenges include environmental conditions (rain, fog, low light), sensor limitations (camera shake, low resolution), and scene complexity (background clutter, occlusion, lighting variations). These disturbances consistently reduce accuracy, causing missed detections, false positives, and localization errors, especially under adverse conditions such as heavy cloud cover (YOLOv8-QSD), low contrast (CDYL), or motion blur (H. Wu et al., 2024). Performance degradation is especially pronounced when small objects suffer from scale changes, irregular dispersion (D. Liao et al., 2025), or dense obstructions, compounded by spatial information loss due to downsampling in remote sensing (C. Chen et al., 2024; Zheng et al., 2024).

To combat noise and input disturbances, models employ specialized architectures such as attention mechanisms (Zheng et al., 2024), transformer hybrids (D. Liao et al., 2025; W. Zhao et al., 2024), and feature refinement modules (Yang et al., 2024; Yuan et al., 2019). However, these often come with computational costs. Heavier parametric structures (D. Liao et al., 2025) and added complexities (Tri-Decoupling++Head) limit deployment on low-resource devices. Lightweight designs (H. Wu et al., 2024) focus on efficiency but remain sensitive to weak lighting. Innovative losses (L. Ni et al., 2024; C. Chen et al., 2024) help improve dense object handling, while strategies like few-shot learning and super-resolution (Yuan et al., 2019) are proposed to enhance robustness. Ultimately, balancing noise resilience with computational efficiency remains crucial, as disturbances increase resource demands and limit real-world applicability.

4. Deep learning techniques for SOD

4.1. Recent trends

Recent trends in SOD focus on overcoming the challenges posed by complex environments through the development of advanced techniques aimed at improving model performance. As shown in Fig. 2, these trends include feature extraction enhancement, feature fusion optimization, attention mechanisms, and advanced learning strategies. In the area of feature extraction, the use of optimized backbone architectures and the design of new feature enhancement modules have led to improved detection accuracy while reducing computational costs. Meanwhile, the effective fusion of features from multiple levels and scales—enabled by networks such as the FPN and lightweight integration modules—has contributed to better SOD performance.

Attention mechanisms also play a crucial role in directing the model's focus toward key regions in the image. These mechanisms—ranging from channel and spatial attention to deformable and transformer-based attention—help suppress irrelevant background

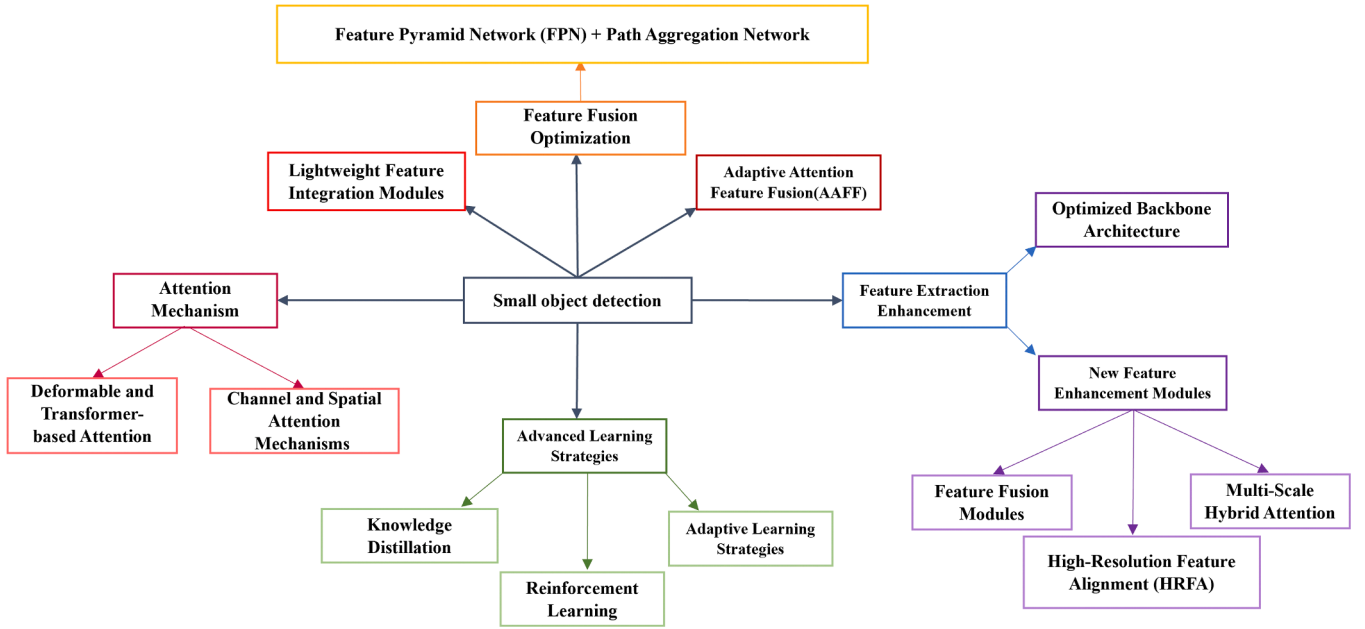


Fig. 2. An overview of recent trends in SOD.

information, thereby enhancing the model's ability to detect small objects. Additionally, advanced learning strategies such as Knowledge Distillation and Reinforcement Learning have improved model adaptability in real-world deployment scenarios.

Considering the introduction and use of recent trends in previous research, as well as the explanation of why these trends are utilized in SOD models, the purpose of presenting Fig. 2 is to provide an overview of these recent trends, introduced collectively. Each trend is then explained in detail, with its application and reasoning also provided.

Table 2 presents the advantages and disadvantages of various approaches in SOD based on the latest trends and techniques (Fig. 2). Each approach has its own strengths and challenges that impact the performance and computational complexity of detection models.

The analysis of recent trends in small object detection reveals that three main approaches—Attention Mechanisms, Feature Fusion Optimization, and Feature Extraction Enhancement—dominate the literature. This indicates a strong focus on improving the model's attention to key regions and effectively integrating multi-scale information. Moreover, Advanced Learning Strategies are increasingly being adopted in real-time applications. The presence of Adaptive Attention Feature Fusion (AAFF) further reflects the shift toward intelligent integration modules in modern architectures. Overall, this distribution highlights a clear move toward models that balance high accuracy, adaptability, and computational efficiency. Fig. 3

To provide a better understanding of the research directions in the field of small object detection, this section presents a detailed analysis of several prominent trends that are broadly outlined in Fig. 2, discussed further in SubSections 4.1.1 to 4.1.6.

4.1.1. Model optimization and lightweight architectures

Recent studies have focused on developing lightweight models tailored for real-time applications in resource-constrained environments. Approaches like modular design, parameter-sharing strategies, and compression techniques have enabled scalable, low-complexity architectures such as FFDet (F. Zhao et al., 2024) and KDSMALL (W. Zhou et al., 2024). These models aim to balance computational efficiency with detection accuracy, ensuring deployment feasibility on drones, mobile devices, and edge computing platforms.

- Example: Integrating SR techniques to enhance image clarity while maintaining lightweight architectures.

4.1.2. Feature fusion and multi-scale detection

Advancements in multi-scale feature extraction and fusion techniques have significantly improved SOD performance. Methods like attention mechanisms, transformer-based detectors (e.g., DETR (DEtection TRansformer) (F. Liu et al., 2025), Swin Transformer), and large kernel designs have enabled better contextual understanding and enhanced detection precision across scales (W. Zhou et al., 2024). Multi-frame feature fusion and cross-modal approaches (Jiang et al., 2024) (e.g., thermal and visible imaging fusion) have further strengthened SOD in complex environments like haze, sandstorms, and maritime settings (W. Zhou et al., 2024; Z. Chen et al., 2024).

- Example: Application of attention-driven fusion to UAV-captured scenarios for improved localization accuracy.

4.1.3. Anchor-free and transformer-based approaches

Anchor-free models have gained attention for their ability to simplify the detection process by directly predicting bounding boxes without predefined anchor points (S. Li et al., 2024). Transformer-based architectures, leveraging self-attention mechanisms, have also shown promise for SOD due to their ability to capture long-range dependencies and multi-scale features effectively.

Anchor-Free and Transformer-based models have emerged as highly promising approaches for SOD due to their ability to simplify the detection process and effectively handle multi-scale features. Anchor-free methods eliminate heuristic anchor design to reduce hyper-parameter sensitivity and memory overhead while improving scalability, while transformer-based approaches leverage self-attention to model long-range dependencies for superior occlusion handling and multi-scale fusion. Notable examples include ESOD and DNTFE-Net, both of which outperform traditional methods in terms of AP, mAP, and FPS, particularly on high-resolution and complex datasets. ESOD shows a notable improvement in accuracy and performance on the VisDrone and TinyPerson datasets, especially in high-resolution images. Similarly, DNTFE-Net delivers excellent results on DIOR and VisDrone, particularly in detecting small objects under complex conditions. These results highlight the advantages of integrating advanced

Table 2

Advantages and disadvantages of various approaches in Fig. 2.

Main Approach Section	Advantages	Disadvantages
Feature Extraction Enhancement	More accurate and meaningful feature extraction using advanced architectures and enhancement modules Increased sensitivity to small objects by aligning high-resolution and multi-scale features	Increased computational load due to the use of complex modules Requires large, high-quality data for effective training
Feature Fusion Optimization	Effective combination of features from different levels with reduced computational cost (e.g., FPN and lightweight modules) Better model focus on key regions using attention mechanisms in the fusion process	Difficulty in synchronizing between highly different scales Potential accuracy loss due to imbalance in information across levels
Attention Mechanisms	Highlighting important regions of the image and suppressing noise and irrelevant backgrounds High flexibility with various types of attention (spatial, channel, adaptive, transformer-based)	Increased computational complexity and processing time Requires specific adjustments and suitable data for optimizing performance
Lightweight Feature Integration Module	- Efficient processing with reduced computational overhead. - Provides a balance between performance and resource consumption, ideal for real-time applications.	- May not capture as rich feature representations as more complex models. - Performance heavily reliant on the integration technique used.
Adaptive Attention Feature Fusion (AAFF)	- Enhanced focus on key features by dynamically adjusting attention based on context. - Improves model performance by integrating multi-scale and high-resolution features more effectively.	- Increased computational complexity due to attention mechanisms. - Potential risk of overfitting if not properly tuned.
Advanced Learning Strategies	Improved model generalization in real-world and diverse conditions using knowledge transfer, multi-step or reinforcement learning Reduced dependence on large labeled datasets	High complexity in implementation and training Requires fine-tuning hyperparameters and significant computational resources

techniques—such as efficient feature extraction—making these models well-suited for SOD in challenging real-world scenarios. Hybrid CNN-transformer models are increasingly explored to harness the advantages of both frameworks (Zhu et al., 2024).

4.1.4. Advanced data augmentation techniques

To mitigate dataset imbalance and enhance generalization, advanced data augmentation methods such as (Generative Adversarial Networks) GAN-based synthetic image generation, hard augmentation strategies, and diffusion models have been widely adopted. These techniques enrich datasets, improve training robustness, and help address the scarcity of small-object-specific samples (Q. Liu et al., 2024).

- Example: Using GANs to generate synthetic UAV datasets for diverse environmental conditions.

4.1.5. Integration of multi-modal and multi-domain data

SOD has increasingly leveraged multi-modal data (e.g., LiDAR (Light Detection and Ranging), radar, infrared, and visual images) to improve

feature extraction and detection accuracy. Combining features across modalities enables the model to process richer representations and handle diverse scenarios effectively. This integration is particularly beneficial in real-world applications like industrial inspections and intelligent traffic systems (R. Jing et al., 2024).

- Example: Fusion of thermal and visible imaging features for industrial inspection tasks.

4.1.6. Knowledge distillation and self-supervised learning

KD has emerged as a key technique for improving the efficiency and real-time performance of SOD models. Additionally, self-supervised learning is being explored to reduce reliance on labeled data, enabling robust model training in unsupervised and semi-supervised scenarios (W. Zhou et al., 2024).

- Example: Deploying KD for lightweight SOD on UAVs.

4.2. Neural network architecture

Table 3 summarizes various papers and methods for SOD, focusing on strategies like feature extraction, feature fusion, and attention mechanisms. Each paper applies different techniques such as motion processing, deformable convolution, RL, and KD to improve SOD accuracy. These methods usually combine multiple approaches to enhance model performance under various conditions and reduce computational costs.

In addition to presenting key innovations related to recent research in Table 3, this paper compares possible training in the presented cases, which are described in detail. Most of these models predominantly rely on supervised training, though with significant variations in initialization and data handling. Several approaches utilize pretrained backbones (e.g., (Lei and Liu, 2024; Jobaer et al., 2025; Y. Zhang et al., 2024)) to facilitate transfer learning, while others, such as MICPL, are trained from scratch (S. Chen et al., 2024) to avoid potential biases introduced by pretrained weights. The nature of the training data significantly influences architectural and training design choices. For instance, models that process low-resolution inputs (e.g., Learning Cross-Modality, Lightweight Fusion Strategy) or operate in dense and cluttered environments (Yang et al., 2024; Ma et al., 2024) often integrate specialized components such as attention mechanisms (e.g., STF-FPN, DSAM) and hierarchical feature fusion modules (e.g., BiFPN, RFP) directly into their training pipelines. Addressing data imbalance is another central concern across many models. The MgD employs a multi-stage sample selection strategy, while SESA-Net implements Sample Importance Assessment—both aiming to prioritize informative samples during training, thereby improving detection robustness and training efficiency.

Advanced training strategies SOD increasingly emphasize loss function innovation and dynamic optimization to address task-specific challenges. Several models employ specialized loss functions tailored to specific challenges. For instance, SESA-Net employs the Lder Loss to (Ma et al., 2024) enhance localization precision for small objects, while Tri-Decoupling++ Head introduces Adaptive Focal-EIoU Loss to mitigate the impact of noisy bounding boxes. Similarly, the Lightweight Fusion Strategy model incorporates Generalized Focal Loss to address feature uncertainty in low-resolution data. Dynamic sampling and spatial adaptation also play a crucial role in recent models. SOD-YOLO integrates DySample, a mechanism that performs adaptive feature upsampling based on object density and background complexity. LA-YOLO leverages a Decoupled Dynamic Head (DD-Head) (Y. Li et al., 2024) to optimize spatial computations during training, enhancing detection efficiency. Furthermore, techniques like multi-task optimization (Yang et al., 2024) and multi-stage training (D. Chen et al., 2024) are employed to jointly optimize classification, localization, and feature suppression, demonstrating a broader trend toward customized training pipelines designed to counter dataset-specific challenges such as

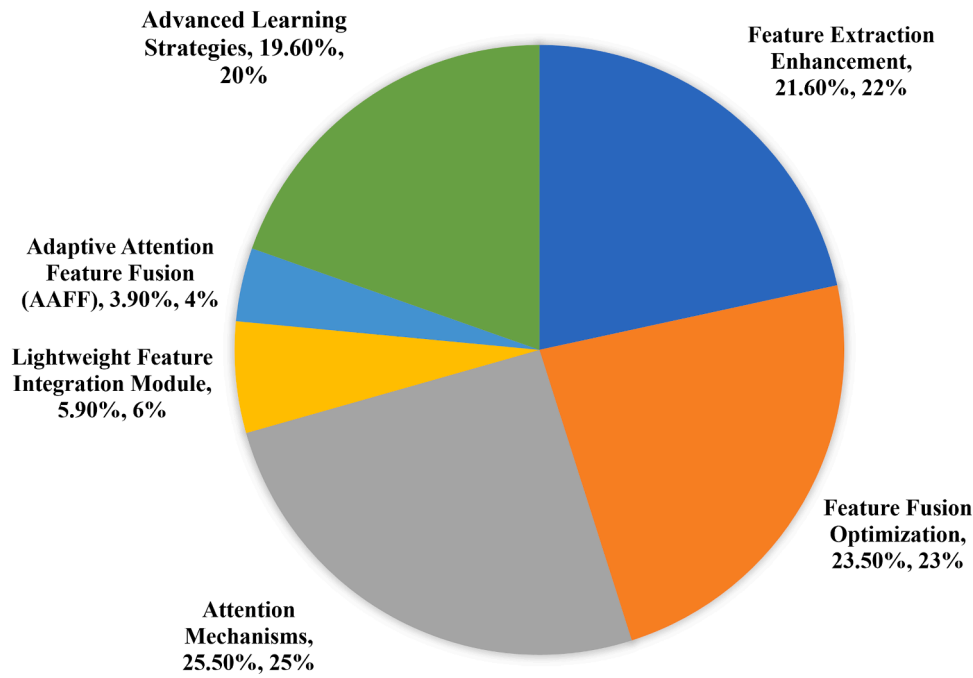


Fig. 3. Distribution of recent trends in SOD papers.

occlusion, scale variation, and background clutter.

4.3. Clarity and visual information improvement

Improving the clarity and visual information in computer vision tasks is crucial for enhancing the performance of models, particularly in areas such as object detection, segmentation, and image classification. One of the key objectives in addressing these challenges is to ensure that models can extract and utilize fine-grained details from images, which is especially important when dealing with small or low-resolution objects. This section explores three main techniques for improving clarity and visual information: SR for image quality enhancement, the use of multi-scale information, and the fusion of information from different layers of neural networks (D. Chen et al., 2024).

4.3.1. SR for image quality enhancement

SR techniques improve the resolution of an image, effectively increasing the clarity of fine details that might otherwise be lost in low-resolution images. SR aims to recover high-resolution details from a series of low-resolution images, typically utilizing deep learning-based models such as CNNs (K. Liu et al., 2025). These models learn to up-scale the low-resolution input into a higher-resolution output, using the contextual information in the image to fill in missing details.

In the context of SOD, SR plays a critical role in improving the quality of images where tiny objects might be indistinguishable due to resolution constraints. By enhancing the resolution of the input image, SR allows models to discern minute details and achieve more accurate localization and classification of small objects. Techniques leveraging GANs for SR have gained significant attention, as GAN-based approaches are capable of generating perceptually realistic high-resolution images by learning the underlying distribution of image data (Y. Zhao et al., 2024).

Moreover, SR methods are often coupled with other image enhancement techniques, such as denoising and deblurring, to ensure that the super-resolved images do not introduce artificial artifacts that could degrade the overall model performance (Jobaer et al., 2025).

4.3.2. Utilizing multi-scale information

Multi-scale information refers to the ability to extract and process

features at various scales from an image, ensuring that both fine-grained details and broader context are captured. Many computer vision tasks, including SOD, benefit from multi-scale feature extraction, as objects can appear at varying scales within images, necessitating information from multiple levels of abstraction for effective detection (Y. Zhao et al., 2024; D. Liao et al., 2025).

Models that incorporate multi-scale information often employ architectures that process images at multiple resolutions or utilize feature pyramid structures to fuse information from different feature map levels (G. Song et al., 2024; B. Wang et al., 2024; F. Feng et al., 2025). This allows the network to recognize both small and large objects by detecting patterns across different scales. For instance, in SOD, fine-grained details at a low resolution (captured by early layers in the network) can be integrated with more abstract, high-level features (extracted by deeper layers), thereby improving the detection of small objects.

A key challenge in utilizing multi-scale information is achieving an effective balance between the spatial and semantic information at each scale. Fine-grained details are often diminished or lost as transitioning to higher resolutions, necessitating architectural designs that can efficiently manage this information flow. Feature pyramids, such as FPN, are commonly employed to address this issue by allowing models to combine high-resolution fine-grained features with coarser contextual information from deeper layers (Z. Chen et al., 2024; Q. Liu et al., 2024; Y. Gao et al., 2024).

4.3.3. Fusion of information across network layers

In modern deep learning architectures, fusing information from different network layers is a crucial technique for improving performance. Feature maps produced at various depths capture distinct aspects of the input image, ranging from low-level edges and textures to high-level semantic information (Zhou and Zhu, 2024; Y. Zhang et al., 2024). Fusing these feature maps allows the network to leverage a more comprehensive understanding of the image, which is essential for tasks such as SOD, where small objects may only be represented in early layers of the network.

Several strategies have been proposed for layer-wise feature fusion, including concatenation, addition, and attention-based mechanisms. Among these, attention mechanisms (Y. Zhang et al., 2024; F. Zhao

Table 3
Key innovations in recent researches on SOD.

Study	Key Contribution	Method	Performance	Limitation	Recent trends Categorization	Technical Justification	Computational Requirement
(B. Wang et al., 2024)	Adaptive lightweight SOD for USV	SHRDet uses HRNet, LASFF, and Focal-EIoU for accuracy, while SHRDet-N adds cross-attention, EWC, and metric loss for few-shot learning.	mAP: 80.7 % on Water Surface Object Detection Dataset (WSODD)	detail loss in deep convolutions; limited USV hardware restricts model complexity; SHRDet-N depends on few-shot sample quality/quantity; minor GPU inefficiencies (multi-branch); dynamic sea surface hampers stable performance; no explicit use of pretrained networks.	Lightweight Network Design: Focus on computational efficiency	SHRDet uses an HRNet backbone with an Enhanced Shuffle Block to cut computation while preserving small object detection performance.	NVIDIA RTX 2080 Ti GPU with 11 GB of VRAM
			—	—		The LASFF module integrates multi-scale features, improving detection accuracy for various object sizes	
			—	—		Focal-EIoU loss enhances accuracy by focusing on harder-to-detect objects in cluttered environments	
			—	—		SHRDet-N uses cross-attention, EWC, and metric loss to adapt to new categories with minimal data in few-shot learning.	
			—	—		Cross-attention and EWC help SHRDet optimize multiple objectives, boosting detection performance.	
(Hao et al., 2024)	Enhances detection accuracy through Slicing-Aided Learning for SOD	combines a Slicing-Aided (SA) Module to divide high-resolution images into patches, a YOLOv5-based detection module (TPH-YOLOv5) with an SPPFCSPC block for faster processing, CARAFE for feature refinement, and a transformer prediction head to improve detection, especially for small objects	mAP50: 67.1 % on VISDRONE2019	false detections remain for dense/overlapping objects; strong dependency on image slicing; overlapping patches + transformer increase memory usage during training; uses pre-trained TPH-YOLOv5 but not explicitly stated for all components.	Integration of Transformer Prediction Head	Enhances detection accuracy, especially for small objects.	NVIDIA RTX3080 GPU
			—	—		Combines YOLOv5-based detection with advanced feature refinement techniques for better performance in small object SOD.	

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Table 3 (continued)

Study	Key Contribution	Method	Performance	Limitation	Recent trends Categorization	Technical Justification	Computational Requirement
(D. Chen et al., 2024)	Enhanced SOD under Sample Imbalance	Incorporates Multi-Granularity Feature Extraction (MFE) using deformable convolution kernels for different object sizes and Sequential Three-Way Selection (S3WS) to address sample imbalance.	MS COCO: APm: 43.7 PASCAL VOC: APm: 76.9 VisDrone-DET 2020: AP50: 65.5	MFE module lacks generalization across tasks; S3WS not optimized for runtime/memory; S3WS performance depends on multiple IoU functions; potential efficiency issues in diverse scenarios.	Feature Extraction Enhancement	The MFE module uses deformable convolutions to capture multi-scale fine-grained features, enhancing SOD accuracy.	Not mentioned
	Lightweight small-defect detection with a novel YOLOv5_ES	The backbone uses SPD-Conv for efficient feature extraction, the neck employs an Efficient Multi-scale Attention (EMA) module for multi-scale fusion, and the detection head focuses on small-scale detection by removing larger scale layers	mAP50: 95.7 on HRPCB	Evaluated only on public datasets; lacks validation with real industrial PCB samples; possible domain shift issues; unclear use of pretrained models; background clutter remains a challenge.	Lightweight Model Design: Optimizing the model for SOD by simplifying the detection head.	The SPD-Conv layer improves fine-grained features while lowering complexity.	NVIDIA RTX4090Ti 24G
(Z. Chen et al., 2024)	Boosting SOD accuracy by using High-Resolution Feature Pyramid Network for Small Object Detection on Drone View.	The HRFA module upsamples multi-scale features, while the HRFF separates and aggregates features for classification and localization. The MSDH handles scale differences by predicting categories and locations at different scales.	VisDrone2019-DET: mAP: 50.8 TINYPERSON: mAP: 52.5	Not effective for large-scale object detection; drone image variations (angles/scales) remain; HR-FPN increases computational load; no pretrained models used; limited to small object detection, tracking extension still future work.	Attention Mechanisms: EMA for multi-scale feature fusion. Custom Convolution Technique: SPD-Conv for enhanced fine-grained feature extraction. Feature Fusion Modules	The EMA module enhances features by capturing multi-scale contextual information. Simplifying the detection head improves detection accuracy for small defects on PCB surfaces. The Feature Fusion Modules boost SOD accuracy by aligning multi-scale features at high resolution, improving scale handling.	12-core processor, 128 G RAM, a RTX 3090 GPU
					High-Resolution Feature Alignment (HRFA) Multi-Scale Hybrid Attention (MHA)	HRFA improves SOD by aligning high-res features for better detail and accuracy. The MHA module improves small object detection by focusing on key features across scales, reducing redundancy, and enhancing detection accuracy.	

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Table 3 (continued)

Study	Key Contribution	Method	Performance	Limitation	Recent trends Categorization	Technical Justification	Computational Requirement
(F. Feng et al., 2025)	Reducing computational complexity with a lightweight small object detection algorithm for UAV aerial imagery.	Adds a small object detection layer (P2) and removes large object layers (P5), introducing dynamic convolution with lightweight modules. Adaptive Multi-Scale Spatial Feature Fusion (AMSFF) integrates high-level semantics and low-level details for better detection.	mAP(0.50): 47.4 on VisDrone2019	Limited computational resources on UAV terminals constrain model complexity; small objects still prone to detection errors and false positives; no mention of pretrained models; needs further accuracy improvement for real-world UAV tasks.	Lightweight Network Design	Reduces computational complexity and enhances small object feature extraction by adjusting the model's depth and width.	GeForce RTX 3090 24 GB graphics memory
			—	—		AMSFF improves feature representation and solves layer information conflict, enhancing detail detection in complex scenarios.	
(Jobaer et al., 2025)	enhancing SOD in blurry environments with UAV assisted images by using a knowledge distillation framework.	Uses a knowledge distillation framework with a deblurring module and attention mechanisms (CBAM for the teacher, SENetV2 for the student) to enhance small object detection in blurry UAV images, based on a modified YOLOv8 architecture.	SOD-Dataset[mAP(0.50): 64.3], VisDrone-Dataset [mAP(0.50): 53.5	Limited UAV computational resources; lack of high-quality training data; current models optimized for clear images, blurry images remain challenging; deblurring adds computational overhead; dataset selection critical for performance.	Knowledge distillation	Teacher-student framework transfers knowledge from a sharp to blurry model, enhancing small object detection in blur.	Two parallel Nvidia RTX 3090Ti 48 GB
			—	—		The attention modules (CBAM for teacher, SENetV2 for student) recalibrate feature maps, enhancing focus on key object features in noisy or blurred images.	
(Zhu et al., 2024)	Small Object Detection Method Based on Global Multi-Level Perception and Dynamic Region Aggregation.	Uses a Global Multi-Level Perception(GMP) module for feature combination and a Dynamic Region Aggregation(DRA) module with sparse attention to improve dense small object detection.	NWPU VHR-10: mAP: 95.5 TT-100K: mAP: 90.8	Computationally complex; high memory requirements; not explicitly using pretrained models; dense object differentiation remains challenging; scalability to larger datasets and complex tasks needs improvement.	Global and local feature fusion	Using GMP module to simultaneously integrate global and local features, resulting in richer feature representations.	Two NVIDIA TITAN RTX GPUs
			—	—		Using the DRA module's sparse attention to highlight key features, reduce noise, and improve dense small object detection.	

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Table 3 (continued)

Study	Key Contribution	Method	Performance	Limitation	Recent trends Categorization	Technical Justification	Computational Requirement
(Y. Zhao et al., 2024)	Degradation-aware dual-branch detection framework in Medium-Low-Resolution Remote Sensing Images.	A dual-branch architecture with an enhancement branch for feature learning in low-resolution images and a detection branch with parallel attention fusion for precise feature extraction and noise suppression.	NWPU VHR-10: mAP: 95.5 TT-100K: mAP: 90.8	Limited evaluation scope; multi-branch design increases computational/training complexity; feature loss during downsampling remains challenging; background clutter affects detection; pretrained model usage not mentioned.	Parallel hybrid attention module	For feature fusion, focusing on targets and suppressing background noise.	NVIDIA3090
	Fine-grained, attention-based multi-scale feature enhancement for SOD.	FFEDet extracts features with DarkNet53 and applies multi-scale spatial and channel attention using the ECFA module. SEConv reduces complexity and increases the receptive field. DFSLoss employs dynamic weighting to focus on hard samples.	PASCAL VOC: [mAP0.5 = 85.5 ± 0.2], VisDrone-DET2021 [mAP0.5 = 53.9 ± 0.2], TGRS-HRRSD [mAP0.5 = 91.4 ± 0.2], DOTAv1. [mAP0.5 = 78.2 ± 0.2]	Complex backgrounds and object scales remain challenging; efficient computation vs feature extraction trade-off; poor-quality samples affect convergence; pretrained model use not clearly mentioned.	Degradation-aware learning Attention Mechanism	Improving feature learning by simulating real-world image degradations in the enhancement branch. The spatial and channel attention mechanisms enable the model to focus on more important regions of the image, enhance features related to small objects, and suppress background noise, thereby improving detection accuracy. Multi-scale feature integration preserves fine details of small objects and combines them with higher-level features, with attention mechanisms enhancing the overall representation quality.	NVIDIA RTX A6000
(S. Chen et al., 2024)	Enhances small moving object detection accuracy through Motion-Inspired Cross-Pattern Learning in satellite videos.	Using the Motion Processing Module (MPSRM) to extract motion features and the Motion-Vision Alignment (MVA) Module to synchronize motion and visual information.	Map:0.87 on DroneCrowd	potential semantic bias in motion patterns; high computational costs limit real-time use; sensitive to training data distribution; not using pretrained models.	Optimized Backbone Architectures	Using MPM to extract motion features and boost small moving object detection accuracy.	Not mentioned
(Tian et al., 2024)	Preserves and fuses key features through object feedback and feature retention for SOD in intelligent transportation scenes.	Uses SOPANet with residual connections to preserve and fuse key features, and Small Object Intersection over Union (SOIoU) Loss to improve localization accuracy.	COCO: mAP50: 62.91 BDD100K: mAP50: 45.28	Standard loss functions not optimized for small objects; multi-scale feature learning reduces crucial details; detectors biased toward large objects; localization accuracy for small objects remains suboptimal.	Feature Fusion Optimization Optimized Backbone Architectures	Using MVA to sync visual and motion data for optimized detection. Employing SOPANet and Residual Connections to preserve key features.	Not mentioned

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Table 3 (continued)

Study	Key Contribution	Method	Performance	Limitation	Recent trends Categorization	Technical Justification	Computational Requirement
(Cao et al., 2024)	Real-time SOD via pillar scoring and self-attention to suppress noise and enhance key cues.	ScorePillar combines pillar-based LiDAR encoding, Point-Score Feature Extraction (PSFE) module, and PillarConv for efficient feature extraction, enhancing real-time small object detection with attention-guided scoring, ResNet backbone, multiscale fusion, and an anchor-based detection head.	—	—	High-Resolution Feature Alignment	Fusing key features with Residual Connections to boost accuracy..	Not mentioned
			—	—	Advanced Learning Strategies	Deploys SOIoU Loss to improve small object localization.	
			mAP(0.50) = 60.29 on KITTI and nuScenes datasets	Sparse point clouds hinder small object detection; Z-axis info loss affects accuracy; limited scalability across environments; computational cost vs real-time trade-off; struggles in dense/occluded scenes; pretrained models not mentioned.	Optimized Backbone Architectures	The model employs a ResNet-based backbone to efficiently extract both fine and coarse spatial features from sparse LiDAR data, which is crucial for capturing small object characteristics like pedestrians and bicycles.	
			—	—	Feature Extraction Enhancement	LiDAR point clouds are encoded into vertical pillars for efficient 2D convolutions, with the PillarConv module using large-kernel convolutions to enhance small object detection.	
(Liu and Jiang, 2024)	Bidirectional adaptive fusion with spatial-aware head for detecting insulator self-explosion small defects.	Uses the Faster-C2f backbone with Bidirectional Adaptive Feature Pyramid Network (Bi-AFPN-P2) for multi-scale fusion and DD-Head for spatial-aware convolutions, improving small object detection in insulator self-explosion defects.	—	—	Attention Mechanisms	Combines self-scoring and mutual-scoring with self-attention mechanism (ECANet) to suppress noise and highlight key small object cues.	Not mentioned
			—	—	Feature Fusion Optimization	The neck module fuses multi-scale features with atrous convolutions, expanding the receptive field without losing resolution, crucial for detecting small objects in sparse LiDAR data.	
			Map:92.51 % on Combines Chinese Power Line Insulator Dataset, TipDMCup challenge data, and self-collected UAV images	Trade-off between lightweight design and high accuracy; real-time performance needs improvement; model adaptability across diverse datasets is limited; pretrained YOLO-based models used but further optimization required.	Optimized Backbone Architectures	The Faster-C2f backbone uses lightweight convolutions to minimize redundancy and enhance feature extraction, particularly for small object detection.	
			—	—	Feature Extraction Enhancement	The Faster-C2f backbone optimizes feature extraction by reducing	

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Table 3 (continued)

Study	Key Contribution	Method	Performance	Limitation	Recent trends Categorization	Technical Justification	Computational Requirement
(Shao et al., 2024)	Dual-path fusion with attention and deformable convolutions for accurate small object detection.	Uses a ResNet backbone for multi-resolution features, a dual top-down pathway for fusion, a Feature-Aware Module (FAM) to reduce the semantic gap, and deformable convolutions in the head to improve detection accuracy for objects of varying shapes and sizes.	mAP50: 71.7 on NEU-DET	Visual similarity of defect types causes false positives; small receptive field limits global info capture; traditional feature fusion loses spatial info; large kernels add computational burden; no pretrained models used.	Optimized Backbone Architectures	parameters and FLOPs, focusing on small objects like insulator self-explosion defects.	Quadro RTX 8000 over 1 GPU
						The Bi-AFPN-p2 neck combines multi-scale features with bidirectional processing for effective small object detection, such as insulator defects.	
						The DD-Head uses spatial-aware convolutions to improve small object localization and detection accuracy for insulator defects by decoupling classification tasks and reducing redundancy.	
						Deploys the ResNet Backbone for multi-resolution feature extraction.	
(J. Wu et al., 2024)	Lightweight federated learning with multilayer fusion for SOD in coal mines.	Utilizing lightweight models and federated networks for multi-layer coordinated learning and optimized SOD, aiming to reduce computational complexity and improve detection accuracy in various conditions.	CMUOD: mAP: 96.2 COCO VAL2017: mAP: 43.5	harsh coal mine environments (low light, dust, vapor) affect accuracy; latency issues in data transmission; concept drift degrades performance; pretrained model usage not explicitly stated.	Optimized Backbone Architectures	Uses a dual top-down pathway for better multi-level feature fusion.	Tesla V100 GPU as servers and use the NVIDIA Jetson TX2 as embedded computing devices (Adapted for EdgeDevice)
						Integrates FAM to reduce the semantic gap and focus on key features.	
						Utilizes deformable convolutions in the Head to enhance the accuracy of detecting small objects with varying shapes and sizes.	
						the deployment of lightweight models to reduce computational complexity and enhance feature extraction efficiency.	

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Table 3 (continued)

Study	Key Contribution	Method	Performance	Limitation	Recent trends Categorization	Technical Justification	Computational Requirement
(Zhou and Zhu, 2024)	Accurate small object detection via balanced label assignment and attention-guided feature alignment.	Integrates spatial and temporal information, enhancing small object features with Foreground Feature Alignment, Background Comparison, and Feature Amalgamation, while reducing noise and improving detection accuracy in side-scan snar images.	Ms COCO: 66.5 DIOR: mAP:73.7 VisDrone: 34.3 56.3 35.2 24.7 46.2 46.1 SODA-D: 30.9 58.3 27.9 12.0 26.9 37.9 48.4	Missed and false detections due to appearance similarities; struggles with texture-blurred objects and limited contextual info; ~50 % miss rate on certain classes; pretrained ResNet-50 used, but discriminative feature extraction still needs improvement.	Feature Fusion Optimization	multi-layer coordination in federated learning for optimal feature fusion.	4 NVIDIA V100 GPUs
					Advanced Learning Strategies	the deployment of federated learning to reduce complexity and improve detection accuracy in various conditions.	
					Feature Extraction Enhancement	Implements Foreground Feature Alignment and Background Comparison	
(S. Li et al., 2024)	Two-stage low-high resolution strategy for accurate, real-time, computationally efficient SOD on FPGA.	A two-stage architecture with low-resolution localization and high-resolution classification to improve accuracy and reduce cost.	TTI00K: mAP: 93.8 GTSDB: mAP: 97.99	Limited accuracy from multi-task learning; small IoU hampers localization; downsampling degrades small object features; FPGA-specific design limits portability; extra steps (upsampling/cropping) add complexity for constrained environments.	Feature Fusion Optimization Attention Mechanisms	Adopts feature amalgamation Uses Channel and Spatial Attention	NVIDIA 3090 (Adapted for EdgeDevice)
					Feature Extraction Enhancement	Integrates a two-stage architecture for hierarchical feature processing	
(Sun et al., 2024)	Enhanced SOD in infrared videos using multi-input fusion and advanced attention mechanisms.	A multi-input module fuses frames and features with optical flow, BiFormer, and GSConv to boost efficiency, accuracy, and small-object detection using α -Wiou v3.	AP50: 0.943 on ISD	Potential challenges in optical flow extraction under low-motion/low-contrast conditions; no explicit mention of limitations in real-world environments like sandstorms or haze.	Feature Fusion Optimization	Combining low- and high-res data to boost accuracy and cut computational costs.	two NVIDIA RTX 3090Ti GPUs
					Advanced Learning Strategies	Using a two-stage adaptive strategy: a low-cost first stage and a high-accuracy second stage.	
					Optimized Backbone Architectures	Adopts a multi-input processing module for extracting spatial and motion features.	
					Multi-Scale Hybrid Attention, High-	Fusing frames with optical flow for better	

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Table 3 (continued)

Study	Key Contribution	Method	Performance	Limitation	Recent trends Categorization	Technical Justification	Computational Requirement
(G. Song et al., 2024)	Improved UAV SOD using multi-scale attention, foreground enhancement, and lightweight attention modules.	Enhancing SOD with MHA, using MsA for cross-scale similarity, Extraction Module (FEM) for foreground focus, and depthwise separable channel attention(DSCA) to reduce redundancy applied post-Neck.	VisDrone2019-DET: AP50: 56.82 UAVDT: AP50: 37.24 CARPK: AP_50: 97.95	Limited labeling of subtle objects; challenges in detecting small objects with low resolution and densely packed objects; complex backgrounds and lighting variability; computational constraints on UAVs.	Resolution Feature Alignment Attention Mechanisms	spatiotemporal processing. Using BiFormer for advanced image feature attention. Using α -WIoU v3 and a detection layer to boost small-object sensitivity. Using MsA to capture cross-scale similarities and refine small-object features.	NVIDIA 2080Ti GPU with 11 GB GPU memory
			—	—	Adaptive Learning Strategies		
			—	—	Optimized Backbone Architectures		
(R. Wang et al., 2024)	Reduces complexity for real-time classroom evaluation using multi-scale fusion and attention-based SOD.	Optimized multi-scale feature fusion with attention for accuracy, reduced complexity for real-time, and combined spatial-temporal processing.	Results are not comparative,rates among different classrooms are just simply reported.	Complex classroom backgrounds and partial occlusion hinder detection; low resolution and contrast challenge small object recognition; real-time analysis imposes computational demands; adaptability and accuracy in specific environments need further improvement.	Multi-Scale Hybrid Attention, High-Resolution Feature Alignment Attention Mechanisms	Using FEM to enhance foreground features and filter noise. Utilizing MHA and DSCA for focused attention on key features. Optimized multi-scale feature extraction for efficient detection of small objects	Not mentioned
			—	—	Optimized Backbone Architectures		
			—	—	Attention Mechanisms	Using attention to improve object differentiation and reduce background noise in SOD.	
(W. Wang et al., 2024)	Improves lightweight SOD using HDPCnv and VOV-GSCSP for efficient feature extraction.	Replacing Halved Deep Pointwise Convolution (HDPCnv) in the Backbone for better features and less complexity, and using View Group Shuffle Cross Stage Partial Network (VOV-GSCSP) in the Neck for lightweight fusion to boost accuracy and speed.	mAP0.5-0.95: 80.8 % ON PASCAL VOC2007	Challenges in accurately detecting small objects with limited pixel information; balancing accuracy and computational efficiency remains difficult; no pretrained models used; further testing needed in diverse environments.	Optimized Backbone Architectures	Utilizing HDPCnv in the Backbone to enhance feature extraction and reduce complexity.	GeForce RTX 3070, 8gb of graphics
			—	—	Multi-Scale Hybrid Attention, High-Resolution Feature Alignment	Applying VOV-GSCSP in the Neck for lightweight feature fusion and accuracy optimization.	
			—	—			

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Table 3 (continued)

Study	Key Contribution	Method	Performance	Limitation	Recent trends Categorization	Technical Justification	Computational Requirement
(Jiang et al., 2024)	Enhance SOD through Multiscale Feature Fusion Network for UAV Aerial Images.	Deploying an optimized Multiscale Feature Fusion Small Object Detection Network (MFSODNet) architecture for small-object detection in UAV images, featuring Multiscale Feature Extraction Module (MSFEM) for rich multiscale extraction, Bidirectional Dense Feature Pyramid Network (BDFPN) for enhanced fusion, and a modified prediction head all integrated into a YOLOv5-based framework for efficient, accurate performance.	—	High computational complexity; real-time performance struggles on resource-constrained UAVs; performance dependent on high-quality datasets; limited generalization to non-UAV applications; lack of focus on energy efficiency; exclusion of transformer-based models; limited interpretability of multiscale fusion process.	Advanced Learning Strategies	Reduction in computational complexity and model optimization for real-time performance	NVIDIA TITAN RTX-24GB
			map@0.85 % on VisDrone and UAVDT benchmark datasets	UAVs; performance dependent on high-quality datasets; limited generalization to non-UAV applications; lack of focus on energy efficiency; exclusion of transformer-based models; limited interpretability of multiscale fusion process.	Optimized Backbone Architectures	MSFEM uses multiple convolution branches to enhance small object features and prevent loss during downsampling.	
			—	—	Feature Extraction Enhancement	MSFEM captures fine details and multiscale features, boosting detection especially for small objects.	
(W. Zhou et al., 2024)	Improved accuracy enabled by a lightweight SOD algorithm based on knowledge distillation.	Employing EfficientNet in the Backbone for multi-scale feature extraction, CBAM in the Neck for spatial and channel attention to boost fusion, and KD in the Head to transfer knowledge from larger to smaller models, enhancing accuracy while keeping efficiency.	Map(0.50) = 0.72 % on PASCAL VOC	Limited feature extraction for small objects; challenges in accurate localization due to size and background complexity; dependency on high-quality datasets; no focus on temporal information; not using pretrained models.	Optimized Backbone Architectures	BDFPN in the Neck fuses shallow and deep features, improving small-object detection in complex, multi-scale scenes.	RTX A4000 * 1 with 16 GB of video memory
			—	—	Feature Fusion Optimization	Leveraging EfficientNet in the Backbone for optimized multi-scale feature extraction.	
			—	—	Multi-Scale Hybrid Attention, High-Resolution Feature Alignment	Deploying CBAM in the Neck for enhanced feature fusion with channel and spatial attention.	
			—	—	Attention Mechanisms	Using CBAM with spatial and channel attention mechanisms.	
			—	—	Advanced Learning Strategies	Applying KD in the Head to boost accuracy and cut computational cost.	

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Table 3 (continued)

Study	Key Contribution	Method	Performance	Limitation	Recent trends Categorization	Technical Justification	Computational Requirement
(Z. Chen et al., 2024)	Optimized ResNet50 with RFRM enhances multi-scale feature extraction for precise frequency-hopping detection in Complex Electromagnetic Interference.	Using an optimized RFRM-CenterNet with ResNet50 Backbone, RFRM for multi-scale feature enhancement, and a specialized Head for precise frequency-hopping signal detection in complex spectrograms.	Map:90 % on simulated dataset	High levels of electromagnetic interference hinder detection; small, transient features of FH signals are difficult to identify; multi-signal environments cause ambiguity; traditional detection methods insufficient for noisy conditions.	Optimized Backbone Architectures	ResNet50 Backbone extracts key spectrogram features, optimized for small object detection in frequency-hopping signals..	Not mentioned
			—	—	Feature Extraction Enhancement	RFRM in the Neck uses convolutions with varied dilation rates to boost multi-scale features and small-object detection. RFRM effectively integrates multi-scale features, improving small-object representation in spectrograms.	
			—	—	Feature Fusion Optimization		
(Q. Liu et al., 2024)	Enhances green crisp plum detection via EMA, AFPN in complex orchard environments.	Leveraging EMA for clear object-background separation, AFPN to preserve low-level features, and minimum point distance intersection over union (MPDIoU) Loss to enhance localization in occlusions.	Map: 89.4 % on self-built green crisp plum dataset	Missed detection for some green crisp plums; variations in lighting/weather (cloudy, rainy) and complex orchard backgrounds affect accuracy; resolution reduction from high-resolution images impacts detection; recall metric could be improved.	Optimized Backbone Architectures	Using AFPN to preserve low-level features and boost feature extraction.	NVIDIA RTX A4000
			—	—	MHA, High-Resolution Feature Alignment	Fusing multi-scale features with AFPN to retain key information.	
			—	—	Attention Mechanisms	Applying EMA for improved object-background separation.	
			—	—	Advanced Learning Strategies	Adopting MPDIoU Loss to improve localization accuracy under occlusion.	
(K. Liu et al., 2025)	Efficient SOD on High-Resolution Images through optimized multi-scale feature extraction.	Optimized multi-scale feature extraction and fusion, attention-based accuracy improvement, reduced complexity for real-time use, and combined spatial-temporal processing.	VisDrone: AP50: 59.7 UAVDT: AP50: 22.5 TinyPerson: APT: 61.3	Potential truncation of larger objects due to patch slicing; reduced global context modeling with Vision Transformers; latency in slicing process; dataset-specific tuning required for optimal performance.	Optimized Backbone Architectures	Optimized multi-scale feature extraction for efficient detection of small objects	two Nvidia V100 GPUs

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Table 3 (continued)

Study	Key Contribution	Method	Performance	Limitation	Recent trends Categorization	Technical Justification	Computational Requirement
(M. Li et al., 2024)	a lightweight SOD model based on multi-dimensional trans-attention module for remote sensing images.	Utilizing multi-head channel and spatial trans-attention (MCSTA) to extract attention features across multiple dimensions (channel and spatial), integrating features in the Neck via PAN, and optimizing performance.	Map:0.86 % on Vis-Drone and PASCAL VOC	—	Attention Mechanisms	Utilizing attention mechanisms to enhance object differentiation and mitigate background noise in SOD.	NVIDIA GTX4090 GPU
				Dependent on training data quality and diversity; computational complexity increases with added modules; optimized for small objects, limiting generalizability to larger objects; deployment on resource-constrained devices still requires further optimization.	Optimized Backbone Architectures	Using MCSTA, a Transformer-based module for multi-dimensional feature extraction.	
					Multi-Scale Hybrid Attention, High-Resolution Feature Alignment	Fusing features with PAN to aggregate information in the Neck.	
(Xiong et al., 2024)	Adaptive fusion and attention enhance UAV target tracking by balancing features and improving localization.	Using Adaptive Feature Fusion Mechanism (AFFM) to balance small and large object features, Soft Pooling to reduce info loss, and Subspace Attention to improve small-object localization while suppressing background noise.	BDD100K: mAP: 56.36 % KITTI: mAP: 93.33 %	Extreme scale imbalance affects detection; small targets easily affected by noise and environmental factors; low resolution leads to information loss; anchor-based YOLOv5 requires significant computation resources; further research needed on anchor-free detection methods.	Optimized Backbone Architectures	Applying MCSTA, which integrates both channel and spatial attention. Using Soft Pooling to reduce information loss during feature extraction.	NVIDIA GeForce RTX 2060 with 12 GB of memory
					Multi-Scale Hybrid Attention, High-Resolution Feature Alignment		
					Attention Mechanisms		
(B. Wang et al., 2024)	Lightweight SOD for USVs using MHA, MsA, FEM, and DSCA for enhanced few-shot detection.	Employing MHA to enhance SOD, incorporating MsA for cross-scale similarity, FEM for foreground feature boost, and DSCA to reduce channel redundancy integrated post-Neck.	mAP: 80.7 % on Water Surface Object Detection Dataset (WSODD)	—	Multi-Scale Hybrid Attention, High-Resolution Feature Alignment	Deploying AFFM for optimal fusion of small and large object features.	
				—	Attention Mechanisms	Employing Subspace Attention to improve spatial localization and reduce background noise.	
				Limited computational power of USVs affects model complexity; performance heavily dependent on few-shot sample quality; minor computational inefficiencies in multi-branch design; no explicit use of pretrained models.	Optimized Backbone Architectures	Applying MsA to extract cross-scale similarities and optimize small object features.	NVIDIA RTX A6000

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Table 3 (continued)

Study	Key Contribution	Method	Performance	Limitation	Recent trends Categorization	Technical Justification	Computational Requirement
(J. Ni et al., 2024)	Improved SOD through an optimized YOLOv8s for UAV Image Scenarios.	Using an optimized YOLOv8s backbone with Scale Compensation	—	—	Multi-Scale Hybrid Attention, High-Resolution Feature Alignment	Employing FEM to enhance foreground features and filter out noise.	Not mentioned
		Feature Pyramid Network (SCFPN) for multi-scale fusion and an ultra-SOD (P2) layer for better small object detection, integrated post-Neck.	—	Trade-off between accuracy and real-time performance; challenges in detecting dense small objects; background noise and occlusion interfere with detection; requires fine-tuning for different datasets; computational efficiency issues remain despite optimizations.	Attention Mechanisms Optimized Backbone Architectures	Leveraging MHA and DSCA for focused attention on key features. Using Convolutional and Bottleneck layers.	
(Butler and Leung, 2024)	Enhances SOD and localization using dual-backbone R-CNN, heatmaps, and inflated IoU training.	Using a dual-backbone with R-CNN and hourglass networks for improved small object proposals, plus objectness heatmaps and inflated IoU training to refine localization integrated post-Neck.	Vehicle-Detection [AP0.50]= 56.9, Aerial-Cars [AP0.50]= 78.9	Challenges in generalization across datasets; background noise and complex environments affect performance; computational complexity despite architecture optimizations; reliance on pre-trained ResNeXt-101 backbone; potential improvements with multi-modal data integration.	Feature Extraction Enhancement Feature Fusion Optimization Adaptive Learning Strategies	Applying SCFPN Employing SCFPN's weighted feature fusion. Using the ultra-small-object detection layer (P2) in the Head. Integrating R-CNN and hourglass backbones to improve small object proposal generation.	not mentioned
					Optimized Backbone Architectures		
					Feature Extraction Enhancement Feature Fusion Optimization Adaptive Learning Strategies	Using objectness heatmaps in the hourglass backbone to refine localization. Modifying FPN to better capture small object features at multiple scales. Inflated IoU Training uses a Gaussian multiplier to boost small-object IoU, improving training stability and accuracy.	

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Table 3 (continued)

Study	Key Contribution	Method	Performance	Limitation	Recent trends Categorization	Technical Justification	Computational Requirement
(H. Wang et al., 2024)	An Improved and robust SOD Algorithm for Autonomous Vehicles	Using enhanced YOLOv8 with Diverse Branch Block (DBB) for scalable extraction, C2f-DBB for small object detection, BiFPN for fusion, Q-block for precise localization, and DyHead for multi-scale	Map:0.34 % on VisDrone Dataset Map:64.5 % on SODA DataSet	Dependence on high-resolution images; sensitivity to environmental factors (rain, fog, low light); computational overhead from additional components; jitter sensitivity in real-world driving; small object dataset imbalance; requires fine-tuning for diverse datasets.	Optimized Backbone Architectures	Enhanced YOLOv8 backbone with C2f modules, decoupled heads, and anchor-free design, optimized for small object detection in autonomous vehicles.	NVIDIA GTX 3090
	leveraging Bidirectional Feature Pyramid Network (BiFPN) and multi-scale attention.	attention—integrated for robust small object detection.					
(Shi et al., 2024)	Enhances SOD in power line inspection using YOLOv8 with MDCM, AAF, and β -Ciou loss.	Using YOLOv8 with Multi-Branch Dilated Convolution Module (MDCM) for multi-scale extraction, Adaptive Attention Feature Fusion (AAFF) for attention-based fusion, and β -Ciou loss for better bounding box regression, enhancing SOD in power line inspection.	mAP= 93.78 % on CCTSDB 2021 dataset	Sparse point cloud representation; voxelization inefficiency for real-time applications; loss of information in Z-axis; limited scalability in dense or diverse environments; tradeoff between computational efficiency and real-time performance.	Feature Extraction Enhancement Feature Fusion Optimization	DBB uses multi-branch training for scalability and single-branch inference to balance complexity and performance for efficient small object detection. BiFPN enhances multi-scale feature fusion, improving small object representation throughout the network. SONet builds on YOLOv8 with MDCM for multi-scale features, boosting small object detection in power line inspection.	NVIDIA GeForce RTX 3090
					Feature Extraction Enhancement Feature Fusion Optimization	MDCM in SONet captures multi-scale features with multi-branch dilated convolutions, improving small object detection in complex power line inspection images. AAF uses attention to adaptively fuse multi-scale features, enhancing focus on small defects in power lines for better detection.	

et al., 2024; Jiang et al., 2024), in particular, have shown promise in dynamically weighting the importance of features from different layers. By assigning more weight to crucial layers or regions in the image, attention mechanisms can help models focus on areas that contribute the most to the task at hand. For example, attention modules like the Spatial Attention Module (SAM) or Channel Attention Module (CAM) enable models to emphasize spatial regions or channels that are most relevant for the detection of small objects.

Another promising approach involves Feature Fusion Networks (FFNs) (Xie et al., 2024), which combine the outputs from different layers or feature maps to create a more holistic representation of the image. By capturing both high-level semantic and low-level spatial information, these networks can improve clarity and help models detect objects more effectively, particularly when those objects are small or poorly represented in individual layers.

4.4. Data augmentation and synthetic data

4.4.1. Data augmentation techniques for SOD

Data augmentation is a widely used strategy to artificially expand the training dataset by applying a variety of transformations to existing images. In the context of SOD, specific augmentation techniques are employed to simulate real-world scenarios and help the model focus on identifying small-scale features (Q. Liu et al., 2024).

- **Rotation and Scaling:** Small objects often appear in varying orientations and sizes. Applying random rotations and scaling adjustments helps models generalize better to different spatial representations of small objects.
- **Cropping and Zooming:** Focused cropping and zooming augmentations ensure that small objects remain within the image frame while forcing the model to learn fine-grained details.
- **Padding and Contextual Cropping:** Augmentations like padding can introduce a buffer around small objects, while contextual cropping emphasizes objects and their surroundings to improve contextual awareness.
- **Brightness, Contrast, and Noise Adjustments:** Adjusting brightness and contrast or adding Gaussian noise ensures that models can handle variations in lighting and image quality, which are common in real-world conditions.
- **Random Erasing and Occlusion Simulation:** Simulating occlusions by partially erasing regions of an image encourage models to learn robust representations and improve detection performance under occluded conditions.

Advanced augmentation methods, such as MixUp, CutMix, and Mosaic, have been particularly impactful for SOD (R. Jing et al., 2024; J. Wu et al., 2024; Y. Zhang et al., 2024). These techniques combine multiple images into a single composite image, creating complex scenarios that enhance the model's ability to discern small objects within cluttered backgrounds.

4.4.2. Synthetic data generation for SOD

While data augmentation transforms existing data, synthetic data generation creates entirely new training samples, addressing the issue of limited datasets for SOD. This approach is particularly advantageous when capturing real-world images of small objects is impractical or expensive (Q. Liu et al., 2024).

- **GANs:** GANs have been extensively used to create high-quality synthetic images with small objects. By training a generator to produce realistic small objects and their backgrounds, GAN-based approaches can augment datasets with rare or diverse object instances (Tian et al., 2024; R. Jing et al., 2024; D. Chen et al., 2024).
- **Computer-Generated Imagery (CGI):** CGI-based synthetic data allows for complete control over the positioning, size, and texture of small

objects in images. This flexibility makes it possible to generate datasets tailored specifically for SOD tasks (Y. Zhang et al., 2024).

- **Simulators and 3D Rendering:** Simulation platforms such as Unity, CARLA, or Blender enable researchers to generate complex environments populated with small objects. These tools allow for adjustments in lighting, camera angles, and occlusions, ensuring that synthetic datasets closely mimic real-world conditions (Tian et al., 2024).

Synthetic datasets (Q. Liu et al., 2024) also provide pixel-perfect annotations for bounding boxes, segmentation masks, and keypoints, which are essential for training detection models. This automated labeling significantly reduces the cost and time associated with manual annotation processes.

4.4.3. Advantages of data augmentation and synthetic data in SOD

The use of data augmentation and synthetic data generation provides several benefits for detecting small objects:

- **Enhanced Generalization:** Models trained on augmented and synthetic datasets are better equipped to generalize to unseen environments, especially when small objects appear in challenging contexts (Q. Liu et al., 2024).
- **Improved Robustness:** Exposure to variations in scale, occlusions, and lighting improves the robustness of detection algorithms (F. Zhao et al., 2024; Q. Liu et al., 2024).
- **Addressing Data Scarcity:** Synthetic data enables the creation of large datasets in domains where real-world data collection is limited, such as aerial imagery or medical imaging (F. Zhao et al., 2024; Q. Liu et al., 2024).
- **Cost Efficiency:** Generating synthetic data is often more cost-effective than collecting and annotating large-scale real-world datasets (F. Zhao et al., 2024; Q. Liu et al., 2024).
- **Class Imbalance Mitigation:** Augmentation and synthetic data generation can be tailored to oversample underrepresented small object classes, ensuring a balanced dataset (F. Zhao et al., 2024).

4.4.4. Challenges and research directions

Despite their benefits, these techniques face certain challenges:

- **Domain Gap:** A significant difference in visual characteristics between synthetic and real-world images can reduce model performance when applied to real-world data. Bridging this gap through domain adaptation techniques is an active area of research (S. Chen et al., 2024).
- **Over-Augmentation:** Excessive or unrealistic augmentations may introduce noise or unrealistic scenarios, negatively impacting model training. Careful tuning of augmentation parameters is crucial (S. Chen et al., 2024; Liang et al., 2020).
- **Scalability of Synthetic Data:** Creating high-quality synthetic datasets with diverse small objects requires significant computational resources and expertise (S. Chen et al., 2024).

Future research directions include exploring automated data augmentation pipelines (e.g., AutoAugment), improving synthetic data realism through advanced generative models (e.g., diffusion models), and incorporating multi-modal synthetic data to enhance contextual learning for SOD.

4.5. Multi-task and transfer learning

4.5.1. Multi-task learning for SOD

Multi-task learning focuses on training a model to perform multiple related tasks simultaneously, enabling the model to share knowledge and features across tasks. For SOD, this approach helps the model learn complementary information from auxiliary tasks, ultimately improving

Table 4
Summary of SOD datasets.

Dataset	Categories	Challenges	Size	Purpose
VisDrone (Zhu et al., 2018)	10 object categories, including pedestrians, bicycles, vehicles, and buses	Varying spatial density, occlusion, environmental factors (weather, lighting), and complex urban and rural backgrounds.	10,209 images in total with 6471 for training, 548 for validation, and 3190 for testing.	Aerial dataset for SOD captured by UAVs, addressing challenges like object size variation, occlusion, and complex backgrounds, with real-world factors such as weather and lighting changes.
DIOR (Li et al., 2020)	20 object categories, including vehicles, pedestrians, and other common objects in urban and rural environments.	Aerial object detection with varying sizes, occlusions, and complex backgrounds.	Large-scale dataset with over 1 million instances.	Focused on object detection in aerial images, particularly for UAV-based applications, offering a diverse set of challenges for robust model training.
DOTA (Ding et al., 2022)	15 object categories, including vehicles, buildings, ships, and planes.	Large-scale aerial images with varying object sizes, complex backgrounds, and occlusions.	2806 high-resolution aerial images with 188,282 object instances.	Aerial dataset with 15 object categories, featuring large-scale images and various challenges like size variation, occlusions, and complex backgrounds. Suitable for training object detection models in satellite and UAV imagery.
VEDAI (Razakarivony and Jurie, 2016)	Primarily focused on vehicle detection, including various vehicle types like cars, trucks, and buses.	Object detection in aerial imagery with varying vehicle sizes, occlusions, and complex backgrounds.	1268 high-resolution aerial images with 2950 vehicle instances.	Designed for vehicle detection in UAV and satellite imagery, providing valuable data for training detection models in diverse real-world environments.
TinyPerson (X. Yu et al., 2020)	Focuses on pedestrian detection, specifically targeting small-sized individuals in complex environments.	Detecting tiny, occluded, and distant pedestrians in crowded and cluttered settings.	72,000 images with annotations for small pedestrian detection.	Designed to address the challenges of detecting small persons in crowded and occluded scenes, useful for applications like surveillance and crowd monitoring.
MS COCO (Lin et al., 2014)	80 object categories, including people, animals, vehicles, and everyday objects.	Detecting objects in complex scenes with varying sizes, occlusions, and background clutter.	Over 330,000 images, with more than 2.5 million object instances annotated.	Designed for object detection, segmentation, and captioning tasks, widely used for training and evaluating models in real-world image analysis.
SODA (Chen et al., 2017)	Focuses on SOD, with annotations for various object types such as vehicles, pedestrians, and others.	Designed for detecting small objects in images, with challenges like occlusion, size variation, and complex backgrounds.	Includes thousands of images with annotated small objects, categorized into subsets based on object size (e.g., extremely small, relatively small).	Aimed at improving SOD, particularly in UAV and aerial imagery, where objects often occupy only a few pixels.
DroneCrowd (Wen et al., 2021)	Focuses on crowd detection and pedestrian tracking in urban environments.	Detecting and counting small, densely packed crowds with occlusions in aerial imagery.	Contains 13,416 annotated objects across several high-resolution aerial images.	Designed for crowd counting and SOD in aerial views from UAVs, useful for surveillance, security, and crowd management applications.
UAVDT (H. Yu et al., 2020)	Includes various object categories such as vehicles, pedestrians, and cyclists.	Object detection and multi-object tracking in aerial images, with challenges like occlusions, scale variations, and complex backgrounds.	50,000 frames across various sequences.	Focuses on object detection and tracking for UAV-captured videos, aimed at developing models for real-time applications like surveillance and traffic monitoring.
BDD100K (Yu et al., 2018)	The dataset includes 10 object categories, such as cars, pedestrians, bicycles, traffic signs, traffic lights, and other driving-related objects.	The challenges in BDD100K include various weather conditions (rain, snow, sunny), lighting variations, occlusion, and the presence of moving objects. Additionally, environmental complexities like urban streets and rural roads add to the challenges.	The dataset consists of 100,000 images captured from cameras mounted on vehicles in various driving conditions.	The purpose of BDD100K is to support research in autonomous driving and object detection in real-world driving scenarios. It is used for training deep learning models in object recognition, traffic sign detection, and simulating various driving conditions.
WSODD (Zhou et al., 2021)	The dataset includes various object categories that might appear on the water surface, such as boats, logs, debris, and other objects floating on water.	Key challenges in WSODD include variations in object size, occlusion due to other objects, reflections on the water surface, and complex environmental conditions like changing water levels, lighting, and weather.	The dataset typically includes a collection of images captured by UAVs, often with annotations of the objects on the water surface. The size of the dataset can vary depending on the source, but it usually contains a large number of images with varying object sizes and conditions.	The purpose of WSODD is to support research in detecting small or floating objects on water surfaces in aerial imagery. This is useful for applications such as monitoring waterways, environmental protection, and detecting hazards or obstacles on water surfaces using UAVs.
AI-TOD (Wang et al., 2020)	AI-TOD typically involves detecting small objects across a variety of categories such as vehicles, animals, pedestrians, and other objects that appear tiny due to their size or distance in the image.	The main challenges in AI-TOD include detecting objects that occupy very few pixels in images, dealing with occlusions, variations in scale, low resolution, and complex backgrounds. Additionally, tiny objects are often difficult to distinguish from noise, making accurate detection harder.	The dataset or approach in AI-TOD usually involves high-resolution images or videos where the objects of interest are much smaller than the average objects in traditional object detection tasks. The datasets used for AI-TOD are often large-scale, involving thousands of images with annotations for tiny objects.	The purpose of AI-TOD is to develop and refine algorithms that can accurately detect and classify very small objects, even in challenging conditions. Applications of AI-TOD include surveillance, remote sensing, medical imaging, and autonomous driving, where detecting tiny objects (e.g., pedestrians, animals, or road hazards) is crucial for safety and decision-making.

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Table 4 (continued)

Dataset	Categories	Challenges	Size	Purpose
KITTI (Liao et al., 2021)	Includes cars, pedestrians, cyclists, and other objects commonly found in urban and rural environments.	Object detection, stereo vision, and tracking in driving scenarios with varying lighting, weather, and motion.	40,000 images, with 20,000 for training and 4000 for testing.	Primarily used for autonomous driving research, providing 3D object annotations and a variety of real-world challenges for detection and tracking tasks.

detection performance. Key multi-task learning strategies are:

- **Shared Feature Representations:** By sharing the backbone of a neural network across tasks such as object detection, segmentation, and classification, the model can learn more robust and generalized features that benefit SOD (M. Li et al., 2024).
- **Auxiliary Tasks:** Tasks like SR, edge detection, and depth estimation are often integrated with SOD to enhance the model’s ability to focus on fine-grained details (L. Zhou et al., 2024). For instance:
 - **SR as an Auxiliary Task:** Generating high-resolution versions of low-resolution input images can improve the visibility of small objects, allowing the model to extract more discriminative features.
 - **Edge Detection:** Learning to identify edges or boundaries can guide the model to locate small objects more precisely.
- **Multi-Scale Learning:** Incorporating tasks at different scales, such as coarse-scale detection for context and fine-scale detection for details, helps improve the detection of small objects.
- **Task-Specific Heads:** Separate output heads for each task allow the model to focus on task-specific objectives while sharing a unified backbone for feature extraction.

Multi-task learning offers significant advantages, particularly for SOD, by leveraging information from multiple tasks to develop enriched feature representations. This enhances the model’s ability to detect subtle and small objects while also acting as a form of regularization to mitigate overfitting through task-related data. Furthermore, it improves computational efficiency by enabling a single model to handle multiple tasks, reducing the resource demands associated with training separate models. However, multi-task learning also presents challenges that must be addressed to ensure optimal performance. Task conflicts may arise when different tasks have competing objectives, potentially leading to suboptimal performance, necessitating effective task balancing strategies such as dynamic loss weighting (S. Li et al., 2024). Additionally, designing appropriate architectures and training strategies becomes increasingly complex when handling diverse tasks, requiring careful model design and optimization techniques.

4.5.2. Transfer learning for SOD

Transfer learning involves reusing a model trained on a large-scale dataset for a new, often smaller, target dataset. This approach is particularly beneficial for SOD, where labeled data is often scarce. Pre-trained models for transfer learning include:

- **Feature Transfer:** Models pre-trained on large datasets such as ImageNet or COCO can provide robust feature extractors that are fine-tuned for SOD tasks. The early layers of such models capture generic features like edges and textures, which are transferable across tasks (Butler and Leung, 2024; Y. Li et al., 2024).
- **Fine-Tuning:** Transfer learning typically involves freezing the lower layers of the pre-trained model and fine-tuning the higher layers on the target dataset to specialize in detecting small objects (Tian et al., 2024; W. Zhou et al., 2024).
- **Domain-Specific Pre-Training:** For certain applications, such as aerial imagery (R. Jing et al., 2024) or medical imaging (Zou et al., 2024), pre-training on a domain-specific dataset further enhances the transferability of features.

Techniques in transfer learning are:

- **KD:** This technique involves transferring knowledge from a large, complex teacher model to a smaller, lightweight student model. KD is particularly effective for SOD when computational resources are limited (W. Zhou et al., 2024; Jobaer et al., 2025).
- **Cross-Domain Transfer:** Models trained on datasets with large objects can be adapted to detect small objects by leveraging techniques like domain adaptation and domain-invariant feature extraction (Cao et al., 2024).
- **Few-Shot Learning:** Transfer learning can be combined with few-shot learning techniques to improve performance in scenarios where only a few labeled examples of small objects are available (B. Wang et al., 2024).

Transfer learning offers several benefits (Butler and Leung, 2024; T. Aibibu et al., 2023), particularly in the context of SOD. It significantly reduces the amount of labeled data required for training, enhancing data efficiency. Additionally, models initialized with pre-trained weights converge faster and demonstrate improved performance compared to models trained from scratch. Transfer learning also improves generalization by leveraging knowledge from large-scale datasets, allowing the model to perform better in unseen scenarios. However, transfer learning comes with its own set of challenges. One major issue is domain shift, where a mismatch between the source and target domains can undermine the effectiveness of transfer learning. Techniques like domain adaptation are necessary to address this problem (Cao et al., 2024). Additionally, fine-tuning on small target datasets may lead to overfitting, particularly when the dataset does not adequately represent the variability of the task.

4.5.3. Integration of multi-task and transfer learning

Combining multi-task learning with transfer learning can amplify the benefits of both approaches. For example, pre-trained multi-task models can be fine-tuned on a specific SOD task, leveraging both shared knowledge across tasks and pre-trained feature representations. This integration provides a powerful framework for handling the challenges of SOD.

The integration of multi-task and transfer learning has demonstrated significant promise in various applications. In aerial imagery, it has proven effective in detecting small vehicles, animals, and buildings from satellite or drone images. In medical imaging, it enhances the identification of small lesions, tumors, or abnormalities in medical scans (C. Chen et al., 2024; W. Zhou et al., 2024). Additionally, in surveillance, it facilitates the detection of small objects such as weapons or unattended bags in crowded scenes (Cao et al., 2024; D. Liao et al., 2025; Z. Chen et al., 2024).

Future research directions include the development of automated task balancing algorithms to dynamically address conflicting objectives in multi-task learning (S. Li et al., 2024). Another key area is domain adaptation, where improving techniques to handle domain shifts in transfer learning will be crucial (Cao et al., 2024). Additionally, exploring the integration of few-shot learning with multi-task learning for SOD presents a promising avenue (Yuan et al., 2019). Lastly, research into continual transfer learning aims to build models capable of adapting to new tasks and domains while retaining knowledge of previously learned tasks.

4.6. Comparative analysis of key advances in model construction, training methodologies, and data preprocessing

Recent advances in SOD stem from significant innovations across three primary domains: model construction, training methodologies, and data preprocessing. Below is a comparative analysis of key advances in each area and their origins:

4.6.1. Advances from model construction (Architectural innovation)

The development of specialized architectures using multi-scale feature fusion, attention mechanisms (channel, spatial, hybrid), and transformer modules enhances feature discrimination for small objects. The introduction of lightweight designs (e.g., efficient backbones, hybrid convolutions) enables deployment on resource-constrained devices (e.g., drones). Novel neck/head designs (e.g., Bi-AFPN-P2 (Liu and Jiang, 2024), DD-Head (Liu and Jiang, 2024), multi-branch heads like STF-FPN (Yang et al., 2024), Tri-Decoupling++Head (Yang et al., 2024)) specifically target improved localization and feature representation for small objects. The integration of boundary-aware mechanisms (e.g., (J. Song et al., 2024)) helps separate foreground objects from complex backgrounds. These structural advances directly improve model capacity to capture subtle features and spatial relationships critical for small objects, leading to higher precision and recall in challenging scenarios (occlusion, clutter).

4.6.2. Advances from training methodologies

The strategic use of transfer learning—leveraging large pre-trained models such as ResNet and CSPDarknet on datasets like ImageNet and COCO—provides robust initial feature extractors. The adoption of advanced loss functions (e.g., Focal Loss variants and IoU-based losses such as Adaptive Focal-EIoU, Lder Loss, and Generalized Focal Loss) addresses class imbalance and improves localization accuracy. The implementation of dynamic sampling strategies (e.g., DySample in SOD-YOLO (J. Song et al., 2024)) and multi-stage sample selection methods (e.g., MgD (J. Song et al., 2024), SESA-Net's Sample Importance Assessment (Ma et al., 2024)) focuses training on hard or informative examples. Knowledge distillation enables the transfer of knowledge from large models to more efficient lightweight ones. Additionally, advanced data augmentation techniques enhance data diversity, particularly for small objects under varying conditions such as lighting and weather. These training advances optimize learning efficiency, mitigate data imbalance issues, improve robustness to noise and variability, and accelerate convergence—ultimately resulting in more reliable and generalizable detectors.

4.6.3. Advances from data preprocessing

The application of image enhancement techniques—such as denoising, normalization, and contrast adjustment—helps mitigate issues caused by low-quality inputs, including blur, noise, and poor lighting. Super-resolution methods are employed to increase the effective resolution of small objects, improving their visibility to detection models. Targeted preprocessing strategies, such as image slicing or patch-based processing, focus computational resources on regions of interest where small objects are likely to appear. In addition, specialized techniques are used to align input data more effectively with model architectures and to compensate for distortions caused by atmospheric or weather conditions. These preprocessing advances directly enhance the quality and suitability of input data, making small objects more discernible and reducing the burden on models to manage extreme noise or low resolution—ultimately improving overall detection performance.

4.6.4. Synthesis

The field progresses through synergistic innovations across all three domains. While architectural advances (Construction) provide the foundational capability to process small objects effectively, training innovations are essential to efficiently and robustly unlock this

potential. Preprocessing techniques serve as critical enablers by ensuring that input data is optimally prepared for the model. Notably, the most impactful breakthroughs often arise from coordinated design across these domains—for example, an attention mechanism (Construction) trained using a specialized loss function (Training) on data enhanced through super-resolution techniques (Preprocessing).

5. Datasets and evaluation metrics

5.1. Datasets

Table 4 summarizes various datasets used for SOD. Each dataset addresses challenges like size variation, occlusions, and complex backgrounds, and is intended for tasks such as surveillance, autonomous driving, and real-time tracking. These datasets, captured from UAVs and satellites, offer valuable data for training models to handle real-world detection challenges.

5.1.1. Aerial image datasets

Aerial imagery datasets are extensively used for SOD due to their real-world relevance and challenging features, including high variability in object sizes, occlusion, and dense clustering.

5.1.1.1. VisDrone. The VisDrone dataset is one of the most widely adopted benchmarks for SOD in aerial imagery (Zhu et al., 2018). Collected by the Machine Learning and Data Mining Lab at Tianjin University, it provides:

- Diversity: 10 object categories, including pedestrians, bicycles, vehicles, and buses, with varying levels of spatial density and occlusion.
- Scale: Over 10,209 images divided into training (6471), validation (548), and testing (3190) subsets.
- Challenges: Captures diverse weather conditions, lighting variations, and complex urban and rural backgrounds.

5.1.1.2. DIOR. The DIOR (Dataset for Object Detection in Optical Remote Sensing Images) dataset (Li et al., 2020) focuses on remote sensing tasks and offers:

- Scale: 23,463 images and 192,472 annotated instances across 20 object categories.
- Variety: Images are collected from multiple sensors under different conditions, making it suitable for testing detection robustness.

5.1.1.3. DOTA. The DOTA (Dataset for Object Detection in Aerial Images) dataset (Ding et al., 2022) is another key resource:

- Multiple Versions: Includes DOTA-v1.0, v1.5, and v2.0, each with incremental improvements in sample size and annotations.
- Features: Over 15 object categories, such as small vehicles, ships, and storage tanks, annotated with rotated bounding boxes for precise localization.

5.1.1.4. VEDAI. The Vehicle Detection in Aerial Imagery (VEDAI) dataset (Razakarivony and Jurie, 2016) is designed for small vehicle detection in aerial images:

- Focus: Emphasizes multi-class detection of vehicles under varying resolutions.
- Challenges: Small object sizes and cluttered environments.

5.1.2. UAV-based datasets

UAVs provide rich datasets for SOD due to their ability to capture images from various altitudes and angles.

5.1.2.1. UAVDT. The Unmanned Aerial Vehicle Detection and Tracking (UAVDT) dataset (H. Yu et al., 2020) focuses on moving object detection:

- **Scale:** Includes over 100 videos with annotations for vehicles under different scenarios like highways and parking lots.
- **Challenges:** Motion blur, small object sizes, and varying weather conditions.

5.1.2.2. SODA-D. The SOD Dataset (SODA-D) (Chen et al., 2017) is specifically designed for detecting small objects captured by UAVs:

- **Features:** Includes diverse weather and lighting conditions to mimic real-world scenarios.
- **Object Categories:** Nine object classes such as pedestrians, cars, and bicycles, annotated with object size categories (large, medium, small, very small).

5.1.2.3. DroneCrowd. This dataset (Wen et al., 2021) targets crowd tracking tasks:

- **Applications:** Useful for detecting and tracking small human objects in crowded urban areas.
- **Characteristics:** High spatial density and occlusion levels make it challenging.

5.1.3. Autonomous driving datasets

Datasets for autonomous driving often include small object annotations for pedestrians, road signs, and vehicles.

5.1.3.1. BDD100K. The Berkeley DeepDrive dataset (Yu et al., 2018) contains 100,000 diverse driving scenes:

- **Focus:** Multi-class detection for objects like traffic lights, pedestrians, and cars.
- **Challenges:** Varying weather, lighting, and urban/rural settings.

5.1.3.2. KITTI. KITTI (Liao et al., 2021) provides annotated datasets for autonomous driving:

- **Features:** Focuses on 3D object detection but includes small object categories like pedestrians and bicycles.

5.1.4. Specialized datasets for SOD

5.1.4.1. TinyPerson. TinyPerson (X. Yu et al., 2020) is specifically designed for detecting small human objects:

- **Scale:** Contains images with high-resolution annotations for small pedestrians.
- **Applications:** Useful for surveillance and crowd analysis.

5.1.4.2. WSODD. The Water Surface Object Detection Dataset (WSODD) (Zhou et al., 2021) targets objects in maritime environments:

- **Focus:** Includes small objects like boats and buoys, often occluded by waves or environmental factors.

5.1.4.3. AI-TOD. The AI-TOD (Artificial Intelligence for Tiny Object Detection) dataset (Wang et al., 2020) includes:

- **Diversity:** Real-world scenarios with annotations for very small objects in aerial imagery.
- **Applications:** Designed for detecting objects like vehicles and ships in challenging environments.

5.1.5. Synthetic datasets

Synthetic datasets (Z. Chen et al., 2024) address the scarcity of labeled data for SOD by simulating realistic conditions.

5.1.5.1. Motion blur synthetic dataset. This dataset introduces motion blur to simulate UAV image capture conditions:

- **Features:** Covers varying levels of motion blur, enabling models to generalize better for blurred real-world images (H. Wu et al., 2024).

5.1.5.2. Self-Collected datasets. Several studies generate synthetic datasets (Zou et al., 2024) for specific tasks, such as detecting industrial components, insulators, or maritime objects:

- **Examples:** Includes datasets for power line insulators, copper wires, and steel bars, annotated under controlled experimental conditions.

5.1.6. Comparative insights and challenges

The reviewed datasets highlight key challenges in SOD:

- **Low Resolution:** Small objects often occupy only a few pixels, making feature extraction difficult (Y. Zhang et al., 2024).
- **Occlusion and Clutter:** Dense spatial arrangements hinder accurate detection (H. Wang et al., 2024).
- **Diverse Conditions:** Variability in weather, lighting, and object orientation adds complexity (Jiang et al., 2024).
- **Domain-Specific Needs:** Different applications require tailored datasets, such as aerial imagery, driving scenarios, or industrial monitoring.

5.1.7. Future directions

For future directions in SOD, expanding datasets to cover under-represented domains is essential for enhancing model robustness. Developing more diverse and large-scale datasets can improve generalization across various scenarios. Additionally, advancements in synthetic data generation, particularly high-fidelity simulations, can supplement real-world datasets, addressing issues of data scarcity. The integration of multi-modal data, such as thermal imaging and LiDAR, offers complementary information that enhances detection accuracy beyond conventional RGB images (Z. Chen et al., 2024; Xu et al., 2024). Furthermore, adopting dynamic annotation techniques, including polygon masks and 3D bounding boxes, can improve the precision of small object labeling, leading to better detection performance in complex environments.

5.2. Evaluation metrics

Evaluating the performance of SOD systems requires specialized metrics to account for the unique challenges posed by small and tiny objects, such as their low resolution, limited features, and high susceptibility to occlusion. This section reviews the most commonly used evaluation metrics in the field of SOD, emphasizing their utility and adaptability across different datasets and scenarios.

5.2.1. Metrics commonly used across general object detection

• Average Precision (AP):

The AP metric measures the area under the Precision-Recall curve for individual classes. It is widely adopted due to its ability to quantify precision across different recall levels. Specific thresholds like AP_{50:50}, AP_{75:75}, and AP_{50:95:50:95} (a COCO-style metric) are often employed, where the subscript indicates the Intersection over Union (IoU) threshold.

- AP₅₀(50): IoU = 0.50
- AP₇₅(75): IoU = 0.75

- AP(50:95): Average AP across IoUs ranging from 0.50 to 0.95 at intervals of 0.05.
- **Mean Average Precision (mAP):**
The mAP aggregates AP values across all classes and is the primary metric for comparing model performance in multi-class object detection. It is particularly useful for datasets with varying object sizes. Metrics such as mAP(50), mAP(75), and mAP(50:95) are frequently reported.

5.2.2. Metrics for size-specific detection

To address the challenges of detecting objects of varying sizes, especially small and tiny ones, metrics are tailored to measure performance across object scales.

- **APS_SS, APM_MM, APL_LL:**
These metrics, used in COCO and similar datasets, evaluate performance for objects categorized by size (C. Chen et al., 2024):
 - APS_SS: For small objects (*area* < (32 × 32) *Pixels*).
 - APM_MM: For medium-sized objects (*area between* (32 × 32) *and* (96 × 96) *Pixels*).
 - APL_LL: For large objects (*area* > (96 × 96) *Pixels*).
- **APT_TT:**
Introduced for datasets like TinyPerson, APT_TT measures AP specifically for tiny objects, which are even smaller than those categorized as "small" in COCO (K. Liu et al., 2025; R. Jing et al., 2024).

5.2.3. Tracking-specific metrics

For SOD in tracking tasks, specialized metrics are used (S. Chen et al., 2024):

- T-AP10, T-AP15, T-AP20:
These metrics evaluate tracking performance at IoU thresholds of 10%, 15%, and 20%, emphasizing the continuity and accuracy of tracking small, often indistinct objects.
- T-mAP:
Similar to mAP but adapted for tracking scenarios, this metric accounts for the temporal consistency of detected objects across frames.

5.2.4. Core metrics for precision and recall

- Precision, Recall, and F1-Score:
These metrics are foundational in SOD, providing insights into a model's ability to minimize false positives (Precision), identify all relevant instances (Recall), and balance the trade-off between the two (F1-Score) (Q. Liu et al., 2024).

5.2.5. Specialized metrics for small object datasets

Certain datasets and tasks define additional metrics tailored to their characteristics:

- Processed Pixel Number Percentage (PPN): This metric, relevant to applications like aerial imagery, measures the proportion of processed pixels during detection, emphasizing computational efficiency (Fang et al., 2024).
- SNR and Coefficient of Variation (CV): Metrics such as SNR and CV evaluate the quality of the detection process, particularly in noisy or low-resolution datasets (Zou et al., 2024).

5.2.6. Dataset-specific metric implementation

- COCO and PASCAL VOC: These datasets employ AP, AP(50), AP(75), and size-specific metrics (APS_SS, APM_MM, APL_LL). The selection of evaluation metrics for SOD is guided by several key considerations. The COCO evaluation protocol offers a more comprehensive

assessment than PASCAL VOC by computing AP across multiple IOU thresholds, whereas VOC relies on a single fixed threshold, which is insufficient for accurately evaluating small objects (Lin et al., 2014). This flexibility allows for a more detailed analysis, especially in SOD. Furthermore, COCO's evaluation criteria are better aligned with the precision requirements in SOD applications, such as surveillance and aerial image analysis, where high positional accuracy is critical (Borji et al., 2015).

- VisDrone and TinyPerson (Z. Chen et al., 2024; K. Liu et al., 2025; R. Jing et al., 2024): Metrics such as APT_TT, mAP, and APS_SS are essential for evaluating models on these datasets due to the predominance of small and tiny objects.
- DIOR and DOTA: These remote sensing datasets utilize mAP, AP(50), AP(75), and APS_SS to assess object detection across complex and diverse scenes (Zhou and Zhu, 2024; Y. Gao et al., 2024; Shi et al., 2024).

Table 5
Future directions in SOD.

Direction	Details
Lightweight and Scalable Networks (J. Liu et al., 2024; D. Liao et al., 2025)	- Enhancing network efficiency by further optimizing lightweight architectures, reducing computational complexity, and improving parameter-sharing mechanisms. - Developing lightweight backbone networks for SOD.
Enhancing Feature Representation (Feng et al., 2024; Xiao et al., 2024)	- Improving multi-scale feature fusion techniques and incorporating dynamic feature focusing networks for better small object representation in complex scenes. - Leveraging SR models for low-resolution environments.
Transformer-Based Innovations (L. Zhou et al., 2024; Jiang et al., 2024)	- Exploring transformer integration to address CNN-based limitations and enhance multi-scale feature processing. - Combining CNN and transformer frameworks to balance efficiency and precision.
Data Augmentation and Domain Adaptation (Cao et al., 2024; Q. Liu et al., 2024; Jobaer et al., 2025)	- Expanding generative techniques like GANs and diffusion models for realistic small-object datasets under varying conditions. - Investigating domain adaptation for robust performance across unseen datasets.
Advanced Label Assignment and Loss Functions (Zhou and Zhu, 2024)	- Optimizing label assignment for imbalanced datasets and small object classes. - Introducing sophisticated loss functions to improve small-object localization and reduce false detection rates.
Cross-Pattern and Temporal Context Integration (B. Zhao et al., 2024)	- Integrating motion pattern mining and temporal context from video sequences to enhance detection of small, fast-moving objects. - Developing algorithms that incorporate temporal information for robust performance.
Applications in Emerging Domains (F. Zhao et al., 2024; J. Ni et al., 2024)	- Expanding research to domains like medical imaging, ocean surveillance, and precision agriculture. - Incorporating SOD models into intelligent systems like SLAM for improved localization accuracy.
Real-Time Performance Optimization (F. Zhao et al., 2024; Jiang et al., 2024; J. Ni et al., 2024; H. Wang et al., 2024)	- Enhancing real-time performance by minimizing overhead via pruning, quantization, and architectural refinements. - Striving for lightweight deployment on resource-constrained devices such as UAVs and IoT sensors.

- Custom Metrics for Specialized Datasets: Some studies develop metrics tailored to their specific datasets, such as datasets with artificially generated motion blur or varying object sizes. For instance, metrics such as Mean Average Recall (mAR) and image quality-based metrics are used for challenging datasets like SODA-D (Zhou and Zhu, 2024; H. Wang et al., 2024; Y. Li et al., 2024) or WSODD.

5.2.7. Computational efficiency metrics

- Frames Per Second (FPS): FPS measures the real-time capability of a detection system, often reported alongside AP or mAP to highlight trade-offs between accuracy and speed (K. Liu et al., 2025).
- Inference Time: The average time required to process an image or a frame, often provided to showcase the practicality of the detection system for resource-constrained applications (Fang et al., 2024).

6. Applications and real-world use cases

SOD has emerged as a critical area of research and development, addressing challenges in detecting objects with minimal size and fine detail in diverse scenarios. Its applications span across numerous domains, leveraging advancements in machine learning and computer vision to solve complex real-world problems. Below is a comprehensive review of the applications and use cases of SOD.

6.1. Remote sensing and aerial surveillance

One of the prominent applications of SOD is in remote sensing and aerial surveillance, where high-resolution satellite and aerial imagery is used for large-scale analysis (Zheng et al., 2024; Y. Li et al., 2024). In urban monitoring, SOD is applied to detect small objects such as vehicles and buildings in complex and densely populated environments. In the context of disaster management, UAVs are used to identify damaged areas and assist in rapid response. In wildlife conservation, UAVs are employed to monitor animal populations and prevent illegal poaching. Additionally, environmental monitoring uses SOD to track changes in vegetation and water resources, and it is also utilized in military reconnaissance to identify enemy assets. In maritime surveillance, SOD helps detect ships and marine vehicles. Moreover, SOD is used to analyze damages caused by natural disasters and to assess natural environments, especially in forested and marine areas. With the development of new techniques like transformers, detection accuracy in complex and crowded environments has been enhanced. These applications have achieved greater success through advanced methods such as SR and improving detection accuracy in cluttered backgrounds.

6.2. UAV and drone applications

UAVs are widely used in SOD due to their high-resolution imaging from various altitudes. UAVs play a crucial role in surveillance and security, enabling real-time monitoring for border patrol, crowd management, and public safety (S. Chen et al., 2024; W. Zhao et al., 2024). In forest fire detection, UAVs help identify small flames and hotspots, providing early warnings for fire management. UAVs are also essential in traffic monitoring, where they track vehicles, bicycles, and pedestrians to analyze traffic flow and manage parking (R. Jing et al., 2024). Additionally, UAV-based search and rescue operations improve efficiency by locating missing persons or vehicles in disaster zones. In agriculture, UAVs monitor crop health, detect pests, and assess irrigation systems, boosting agricultural productivity (C. Chen et al., 2024). Recent advances, including the use of transformers and SR techniques, have enhanced SOD performance, especially in complex environments like agricultural fields and disaster zones. These technologies have

improved the detection of small objects in cluttered and dynamic settings, making UAVs an indispensable tool in various applications.

6.3. Autonomous systems

SOD plays a vital role in various fields such as autonomous driving, detecting small-scale traffic signs, pedestrians, bicycles, and obstacles to ensure road safety. In autonomous driving, detecting small-scale traffic signs, pedestrians, bicycles, and obstacles at distances is crucial for road safety (J. Liu et al., 2024). Additionally, SOD helps autonomous navigation by assisting USVs and autonomous drones in path planning and obstacle avoidance in complex environments (B. Wang et al., 2024). Furthermore, in Simultaneous Localization and Mapping (SLAM), detecting small objects significantly improves mapping and navigation accuracy, enhancing the capabilities of robotic systems in dynamic environments.

6.4. Industrial applications

In industrial settings, SOD plays a critical role in ensuring the quality and safety of manufacturing processes and products. It is used in defect detection, where subtle surface defects on mechanical equipment, PCBs, and vehicle components are identified to maintain product integrity (Xu et al., 2024). For example, in PCB inspection, SOD methods are employed to detect micro-scale defects such as cracks, scratches, or misalignments that are not visible to the human eye, ensuring the reliability of electronic devices. Additionally, SOD aids in aerospace inspection, helping to detect imperfections in aircraft structures and components. This is particularly important for identifying small cracks in structural elements that could lead to catastrophic failures if left undetected. In cable manufacturing, this technology is utilized to monitor the production process for defects or inconsistencies, ensuring high-quality outputs. A recent case study in cable production utilized UAVs equipped with SOD to inspect cables for micro-cracks during manufacturing, significantly improving detection rates and reducing the need for manual inspection.

6.5. Surveillance and security

SOD significantly enhances security systems by identifying subtle threats or anomalies in real-time. In video surveillance (Jain et al., 2024), it helps detect intruders, suspicious objects, or abnormal activities in high-resolution CCTV footage. For border security, SOD is employed to monitor small-scale movements or objects across borders using drones or stationary sensors, improving surveillance efficiency and response times. For industrial inspections, SOD plays a vital role in identifying small-scale defects in manufacturing environments. For example, in PCB inspection, SOD methods can detect micro-scale defects such as cracks, scratches, or misalignments that may not be visible to the naked eye, ensuring the reliability of electronic devices. Additionally, SOD aids in aerospace inspection, helping to detect imperfections in aircraft structures and components, especially small cracks that could lead to catastrophic failures if left unnoticed. In cable manufacturing, UAVs equipped with SOD have been used to inspect cables for micro-cracks during the manufacturing process, improving defect detection rates and reducing manual inspection efforts. This integration of SOD into industrial settings ensures the safety and quality of products, particularly in complex or high-risk environments (F. Zhao et al., 2024).

6.6. Environmental and natural resource monitoring

SOD plays a key role in sustainable environmental management and resource monitoring. For example, in natural resource surveys, SOD is used to detect small-scale changes such as mineral deposits and water levels. In forestry management, it aids in monitoring tree health, detecting illegal logging, and tracking wildlife movement, especially

using UAVs equipped with SOD for precise, real-time data collection. Additionally, SOD is critical in geological mapping for identifying subtle features like faults, improving exploration efforts (Feng et al., 2024). Recent applications also include monitoring water bodies for changes in quality or pollutants, enhancing resource management.

6.7. Medical imaging and biological analysis

SOD is crucial in healthcare for detecting small-scale abnormalities, such as early tumors or strokes, in diagnostic images, aiding early diagnosis and treatment (Zou et al., 2024). In biological research, SOD helps monitor biological tissues and cellular structures, enhancing diagnostic accuracy. For example, it's applied to detect micro-tumors in mammography, improving disease detection in its early stages. This method is also valuable for analyzing cell division and tissue damage in biological studies.

6.8. Maritime and oceanographic applications

SOD in the maritime and oceanographic domain has extensive applications in monitoring and navigation in challenging environments. In particular, in autonomous ship navigation, SOD is used to identify obstacles such as buoys, debris, or other vessels, ensuring safe navigation. In marine life research, SOD is utilized to track marine organisms, underwater features, and small-scale oceanographic changes, providing valuable insights into the marine ecosystem (Shao et al., 2024). Additionally, in security patrols, SOD is employed to monitor coastal and offshore areas, helping to identify illegal activities or potential threats, which enhances maritime security.

In this regard, the use of UAVs and drones equipped with SOD for more precise detection of obstacles and threats in various environments is on the rise, and this technology is expected to play a significant role in maritime security in the future.

6.9. Intelligent transportation systems

SOD plays a key role in enhancing the efficiency and safety of transportation systems. In traffic sign detection, it helps recognize small traffic signs under various weather and lighting conditions, which is essential for the operation of autonomous vehicles (Shao et al., 2024). For pedestrian flow monitoring, SOD is used to estimate pedestrian density and movement, aiding in urban traffic management and improving public safety (D. Liao et al., 2025). Additionally, in railway and roadway inspections, SOD is used to detect minor faults or obstacles on tracks and roads, contributing to the maintenance and safety of transportation infrastructure.

6.10. Specialized applications in agriculture and forestry

SOD plays a pivotal role in precision agriculture (Vaghela et al., 2025) and forestry management by enabling targeted monitoring. In smart agriculture, it aids in the identification of pests, small weeds, or crop diseases in real-time, allowing for timely interventions to optimize crop yield and health. For instance, in agricultural drone imagery, SOD can detect small-scale pests on crops, such as aphids, which are hard to identify with traditional methods. By using drones equipped with SOD techniques, farmers can quickly respond to infestations, reducing the use of pesticides and optimizing yield. Similarly, in forest monitoring, SOD is used to detect illegal logging activities, track wildlife movement, and monitor changes in forest health from aerial imagery, supporting effective conservation and management efforts. In particular, UAVs equipped with SOD have been shown to successfully identify small logging sites in dense forests, significantly improving forest preservation efforts. These applications contribute to enhanced decision-making and efficiency in resource management, promoting sustainability in both agriculture and forestry (Zheng et al., 2024).

6.11. Comparative analysis and real-world implementation considerations

While the applications of SOD span a wide array of domains, they exhibit differing operational contexts, technical requirements, and deployment challenges. For instance, real-time applications such as autonomous driving and UAV surveillance demand low-latency and high-precision SOD models that can function under resource-constrained environments, whereas medical imaging and industrial inspection prioritize detection accuracy and robustness over real-time performance. Additionally, the complexity of object appearance (e.g., occlusion, scale variation, cluttered backgrounds) varies significantly across domains, affecting model generalizability and the choice of architecture.

In real-world deployments, SOD models are often integrated into larger systems—such as embedded hardware on drones, edge-computing devices in smart cities, or multi-modal diagnostic platforms in healthcare—requiring not only model efficiency but also interoperability with other sensing and decision-making modules. For example, in UAV-based wildfire detection, SOD outputs may be fed into geospatial mapping systems to trigger early warning alerts. Similarly, in autonomous vehicles, SOD assists object recognition pipelines by filtering scene elements for further semantic interpretation.

Expectations for real-world use of SOD models include high detection fidelity under diverse environmental conditions, adaptability to unseen data, and efficient computation on edge devices. However, challenges such as domain shift, limited annotated datasets for specific use cases, and interpretability of results remain open research problems. To advance the practical deployment of SOD, future work should emphasize domain-specific model tuning, real-time optimization, and integration with complementary AI modules (e.g., object tracking, classification, and decision support).

7. Future directions

Recent advancements in SOD have focused on several key areas to enhance performance and deployment feasibility across diverse platforms and environments (Table 5). One area of development is lightweight and scalable networks, where efforts are directed at optimizing architectures to reduce computational complexity and improve parameter-sharing mechanisms, alongside developing lightweight backbone networks specifically tailored for SOD (J. Liu et al., 2024; D. Liao et al., 2025). Additionally, feature representation has been enhanced through multi-scale feature fusion techniques and the incorporation of advanced modules like dynamic feature focusing networks, improving the detection of small objects in cluttered scenes. Leveraging SR models has also shown promise in extracting fine-grained features in low-resolution or degraded environments (García-Murillo et al., 2023; Xiao et al., 2024). In terms of transformer-based innovations, the integration of transformers into SOD models addresses limitations of CNN-based architectures, particularly in multi-scale feature processing, and combining CNN and transformer frameworks offers a balance of efficiency and precision (L. Zhou et al., 2024; Jiang et al., 2024).

Further advancements are seen in data augmentation and domain adaptation, where generative techniques like GANs and diffusion models are used to create realistic small-object datasets under varying environmental conditions, while domain adaptation techniques ensure robust detection performance across unseen datasets and challenging scenarios (Cao et al., 2024; Q. Liu et al., 2024; Jobaer et al., 2025). Advanced label assignment and loss functions are also a focus, with efforts to optimize label assignment methods for imbalanced datasets and small object classes, alongside introducing sophisticated loss functions to improve small-object localization and reduce false detection rates (Zhou and Zhu, 2024).

To address the detection of small, fast-moving objects, cross-pattern and temporal context integration is being explored. This includes

integrating motion pattern mining and temporal context from video sequences, as well as developing algorithms that incorporate temporal information for improved performance in dynamic environments (B. Zhao et al., 2024). Furthermore, applications in emerging domains are being expanded, with research exploring new fields such as medical imaging, ocean surveillance, and precision agriculture, as well as incorporating SOD models into intelligent systems like SLAM for improved localization accuracy in real-world applications (F. Zhao et al., 2024; J. Ni et al., 2024). Finally, real-time performance optimization focuses on minimizing computational overhead through pruning, quantization, and architectural refinements, aiming for lightweight deployment on resource-constrained devices such as UAVs, industrial robots, and IoT sensors (F. Zhao et al., 2024; Jiang et al., 2024; J. Ni et al., 2024; H. Wang et al., 2024).

Looking ahead, both lightweight networks and transformer-based models are anticipated to play significant roles in the advancement of SOD, but their impact will largely depend on the specific use case and available resources. Lightweight networks become especially important when used in edge devices because these devices often have limited resources and require models with lower computational demands. In such cases, reducing computational costs is critical. On the other hand, when accuracy is the primary priority and powerful servers with high computational resources are available, transformer-based models can offer higher accuracy. These models excel at capturing long-range dependencies and multi-scale features, offering superior performance in tasks that demand high precision.

Both lightweight networks and transformer-based models represent promising future directions in SOD. The selection between these approaches is primarily influenced by the trade-off between computational efficiency and accuracy. Transformer-based models are preferable when high accuracy and the ability to process complex, large-scale data are required—particularly in scenarios where computational resources are not a limiting factor. Conversely, lightweight networks are more suitable for resource-constrained environments, such as edge devices, where low latency and reduced computational overhead are critical.

8. Conclusion

SOD remains a vital yet challenging task in computer vision, with broad applications spanning surveillance, autonomous systems, medical imaging, and remote sensing. The unique challenges posed by small objects—such as limited resolution, susceptibility to background interference, and class imbalance—have driven innovation in advanced deep learning techniques including multi-scale feature extraction, attention mechanisms, and super-resolution approaches. The development of specialized datasets and robust evaluation metrics has further propelled the field by enabling more accurate benchmarking and model improvement. Recent trends such as lightweight architectures, transformer-based models, and knowledge distillation show great promise in enhancing efficiency and scalability. Additionally, the integration of multi-modal data and domain-specific strategies continues to expand the practical applications of SOD in increasingly complex and diverse environments.

Looking forward, advancing this field will hinge on addressing three key research areas:

- Domain adaptation and generalization: Developing models that perform reliably across diverse domains and datasets without requiring extensive re-labeling.
- Real-time processing with high efficiency: Designing lightweight and fast architectures suitable for practical, real-world applications.
- Improving accuracy under challenging conditions: Enhancing model robustness in scenarios involving severe occlusion, complex backgrounds, and high noise levels.

By exploring innovative methodologies and leveraging

interdisciplinary insights, SOD is poised to play a transformative role in solving critical industrial problems and ensuring the reliability of intelligent systems in demanding environments.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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